复旦数学资料汇编
金融中的随机分析

来源：香蕉空间 Solution 资料库

2026 年 10 月 13 日

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第一章　金融中的随机分析

1.1　笔记: 金融中的随机分析

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We make the following assumptions throughout.

- There is a probability triple $(\Omega, \mathcal{F}, P)$ where
  - P is the "true" or physical probability measure
  - Ω is the universe of possible outcomes. We use $\omega \in \Omega$ to represent a generic outcome, typically a sample path(s) of a stochastic process(es).
  - the set $\mathcal{F}$ represents the set of possible events where an event is a subset of Ω.
- There is also a filtration, $\{\mathcal{F}_t\}_{t\geq 0}$, that models the evolution of information through time. So for example, if it is known by time t whether or not an event, E, has occurred, then we have $E \in \mathcal{F}_t$. If we are working with a finite horizon, $[0, T]$, then we can take $\mathcal{F} = \mathcal{F}_T$.
- We also say that a stochastic process, X_t, is $\mathcal{F}_t$-adapted if the value of X_t is known at time t when the information represented by $\mathcal{F}_t$ is known. All the processes we consider will be $\mathcal{F}_t$-adapted so we will not bother to state this in the sequel.
- In the continuous-time models that we will study, it will be understood that the filtration $\{\mathcal{F}_t\}_{t\geq 0}$ will be the filtration generated by the stochastic processes (usually a Brownian motion, W_t) that are specified in the model description.

1.2　Martingales and Brownian Motion

Definition 1.2.1. *A stochastic process, $\{W_t : 0 \leq t \leq \infty\}$, is a standard Brownian motion if*

1. $W_0 = 0$
2. *It has continuous sample paths*
3. *It has independent, normally-distributed increments.*

Definition 1.2.2. *An n-dimensional process, $W_t = (W_t^{(1)}, \ldots, W_t^{(n)})$, is a standard n-dimensional Brownian motion if each $W_t^{(i)}$ is a standard Brownian motion and the $W_t^{(i)}$'s are independent of each other.*

Definition 1.2.3. *A stochastic process, $\{X_t : 0 \leq t \leq \infty\}$, is a martingale with respect to the filtration, $\mathcal{F}_t$, and probability measure, P, if*

1. $\mathrm{E}^P||X_t|| < \infty$ *for all* $t \geq 0$
2. $\mathrm{E}^P[X_{t+s}|\mathcal{F}_t] = X_t$ *for all* $t, s \geq 0$.

Example 1.2.1. *(Brownian martingales) Let W_t be a Brownian motion. Then W_t , $W_t^2 - t$ and* $\exp\left(\theta W_t - \theta^2 t/2\right)$ *are all martingales.*

The latter martingale is an example of an exponential martingale. Exponential martingales are of particular significance since they are positive and may be used to define new probability measures.

Example 1.2.2. *(Conditional expectations as martingales) Let Z be a random variable and set $X_t :=$* $\mathrm{E}[Z|\mathcal{F}_t]$. *Show that X_t is a martingale.*

1.3 Quadratic Variation

Consider a partition of the time interval, $[0, T]$ given by

$$0 = t_0 < t_1 < t_2 < \ldots < t_n = T.$$

Let X_t be a Brownian motion and consider the sum of squared changes

$$Q_n(T) := \sum_{i=1}^{n} [\Delta X_{t_i}]^2$$

where $\Delta X_{t_i} := X_{t_i} - X_{t_{i-1}}$.

Definition 1.3.1. *(Quadratic Variation) The quadratic variation of a stochastic process, X_t, is equal to the limit of $Q_n(T)$ as $\Delta t := \max_i(t_i - t_{i-1}) \to 0$.*

Theorem 1.3.1. *The quadratic variation of a Brownian motion is equal to T with probability* 1.

The functions with which you are normally familiar, e.g. continuous differentiable functions, have quadratic variation equal to zero. Note that any continuous stochastic process or function that has non-zero quadratic variation must have infinite **total variation** where the total variation of a process, X_t, on $[0, T]$ is defined as

$$\text{Total Variation} := \lim_{\Delta t \to 0} \sum_{i=1}^{n} |X_{t_k} - X_{t_{k-1}}|.$$

This follows by observing that

$$\sum_{i=1}^{n} (X_{t_k} - X_{t_{k-1}})^2 \leq \left(\sum_{i=1}^{n} |X_{t_k} - X_{t_{k-1}}| \right) \max_{1 \leq k \leq n} |X_{t_k} - X_{t_{k-1}}|. \tag{1}$$

If we now let $n \to \infty$ in (1) then the continuity of X_t implies the impossibility of the process having finite total variation and non-zero quadratic variation. Theorem 2.2 therefore implies that the total variation of a Brownian motion is infinite. We have the following important result which will prove very useful when we price options when there are multiple underlying Brownian motions, as is the case with quanto options for example.

Theorem 1.3.2. *(Lévy's Theorem) A continuous martingale is a Brownian motion if and only if its quadratic variation over each interval $[0, t]$ is equal to t.*

Another interesting result is the following:

Theorem 1.3.3. *Any non-constant continuous martingale must have infinite total variation.*

We know from our discrete-time models that any arbitrage-free model must have an equivalent martingale measure. The same is true in continuous-time models. Theorem 2.4 then implies that if we work in continuous-time with continuous (deflated) price processes, then these processes must have infinite total variation.

1.4 Stochastic Integrals

We now discuss the concept of a stochastic integral, ignoring the various technical conditions that are required to make our definitions rigorous. In this section, we write $X_t(\omega)$ instead of the usual X_t to emphasize that the quantities in question are stochastic.

Definition 1.4.1. *A stopping time of the filtration $\mathcal{F}_t$ is a random time, τ, such that the event $\{\tau \leq t\} \in \mathcal{F}_t$ for all $t > 0$.*

In non-mathematical terms, we see that a stopping time is a random time whose value is part of the information accumulated by that time.

Definition 1.4.2. *We say a process, $h_t(\omega)$, is elementary if it is piece-wise constant so that there exists a sequence of stopping times $0 = t_0 < t_1 < \ \ldots \ < t_n = T$ and a set of $\mathcal{F}_{t_i}$-measurable functions, $e_i(\omega)$, such that*

$$h_t(\omega) = \sum_i e_i(\omega) I_{[t_i, t_{i+1})}(t)$$

where $I_{[t_i,t_{i+1})}(t) = 1$ if $t \in [t_i, t_{i+1})$ and 0 otherwise.

Definition 1.4.3. *The stochastic integral of an elementary function, $h_t(\omega)$, with respect to a Brownian motion, W_t, is defined as*

$$\int_0^T h_t(\omega) dW_t(\omega) := \sum_{i=0}^{n-1} e_i(\omega)(W_{t_{i+1}}(\omega) - W_{t_i}(\omega)). \tag{2}$$

Note that our definition of an elementary function assumes that the function, $h_t(\omega)$, is evaluated at the **left-hand point** of the interval in which t falls. This is a key component in the definition of the stochastic integral: without it the results below would no longer hold. Moreover, defining the stochastic integral in this way makes the resulting theory suitable for financial applications. In particular, if we interpret $h_t(\omega)$ as a trading strategy and the stochastic integral as the gains or losses from this trading strategy, then evaluating $h_t(\omega)$ at the left-hand point is equivalent to imposing the *non-anticipativity* of the trading strategy, a property that we always wish to impose.

For a more general process, $X_t(\omega)$, we have

$$\int_0^T X_t(\omega) dW_t(\omega) := \lim_{n\to\infty} \int_0^T X_t^{(n)}(\omega) dW_t(\omega)$$

where $X_t^{(n)}$ is a sequence of elementary processes that converges (in an appropriate manner) to X_t.

Example 1.4.1. *We want to compute* $\int_0^T W_t dW_t$. *Towards this end, let* $0 = t_0^n < t_1^n < t_2^n < \ldots < t_n^n = T$ *be a partition of* $[0, T]$ *and define*

$$X_t^n := \sum_{i=0}^{n-1} W_{t_i^n} I_{[t_i^n, t_{i+1}^n)}(t)$$

where $I_{[t_i^n, t_{i+1}^n)}(t) = 1$ *if* $t \in [t_i^n, t_{i+1}^n)$ *and is* 0 *otherwise. Then* X_t^n *is an adapted elementary process and, by continuity of Brownian motion, satisfies* $\lim_{n\to\infty} X_t^n = W_t$ *almost surely as* $\max_i |t_{i+1}^n - t_i^n| \to 0$. *The Itô integral of* X_t^n *is given by*

$$\begin{aligned}\int_0^T X_t^n dW_t &= \sum_{i=0}^{n-1} W_{t_i^n}(W_{t_{i+1}^n} - W_{t_i^n}) \\ &= \frac{1}{2}\sum_{i=0}^{n-1}\left(W_{t_{i+1}^n}^2 - W_{t_i^n}^2 - (W_{t_{i+1}^n} - W_{t_i^n})^2\right) \\ &= \frac{1}{2}W_T^2 - \frac{1}{2}W_0^2 - \frac{1}{2}\sum_{i=0}^{n-1}(W_{t_{i+1}^n} - W_{t_i^n})^2. \end{aligned} \tag{3}$$

By the definition of quadratic variation the sum on the right-hand-side of (3) converges in probability to T. *And since* $W_0 = 0$ *we obtain*

$$\int_0^T W_t dW_t = \lim_{n\to\infty}\int_0^T X_t^n dW_t = \frac{1}{2}W_T^2 - \frac{1}{2}T.$$

Note that we will generally evaluate stochastic integrals using Itô's Lemma (to be discussed later) without having to take limits of elementary processes as we did.

Definition 1.4.4. *We define the space* $L^2[0, T]$ *to be the space of processes,* $X_t(\omega)$, *such that*

$$\mathrm{E}\left[\int_0^T X_t(\omega)^2 dt\right] < \infty.$$

Theorem 1.4.1. *(Itô's Isometry) For any* $X_t(\omega) \in L^2[0, T]$ *we have*

$$\mathrm{E}\left[\left(\int_0^T X_t(\omega) dW_t(\omega)\right)^2\right] = \mathrm{E}\left[\int_0^T X_t(\omega)^2 dt\right].$$

证明. (For the case where X_t is an elementary process) Let $X_t = \sum_i e_i(\omega) I_{[t_i, t_{i+1})}(t)$ be an elementary process where the $e_i(\omega)$'s and t_i's are as defined in Definition 3.2. We therefore have $\int_0^T X_t(\omega) dW_t(\omega) := \sum_{i=0}^{n-1} e_i(\omega)(W_{t_{i+1}}(\omega) - W_{t_i}(\omega))$. We then have

$$\begin{aligned}&\mathrm{E}\left[\left(\int_0^T X_t(\omega) dW_t(\omega)\right)^2\right] \\ &= \mathrm{E}\left[\left(\sum_{i=0}^{n-1} e_i(\omega)(W_{t_{i+1}}(\omega) - W_{t_i}(\omega))\right)^2\right] \\ &= \sum_{i=0}^{n-1} \mathrm{E}\left[e_i^2(\omega)(W_{t_{i+1}}(\omega) - W_{t_i}(\omega))^2\right] \\ &\quad + 2\sum_{0\le i<j\le n-1} \mathrm{E}\left[e_i e_j(\omega)(W_{t_{i+1}}(\omega) - W_{t_i}(\omega))(W_{t_{j+1}}(\omega) - W_{t_j}(\omega))\right]\end{aligned}$$

$$
\begin{aligned}
&= \sum_{i=0}^{n-1} \mathrm{E}\left[e_i^2(\omega) \underbrace{\mathrm{E}_{t_i}\left[(W_{t_{i+1}}(\omega) - W_{t_i}(\omega))^2 \right]}_{=t_{i+1}-t_i} \right] \\
&\quad + 2 \sum_{0 \le i < j \le n-1} \mathrm{E}\left[e_i e_j (\omega)(W_{t_{i+1}}(\omega) - W_{t_i}(\omega)) \underbrace{\mathrm{E}_{t_j}\left[(W_{t_{j+1}}(\omega) - W_{t_j}(\omega)) \right]}_{=0} \right] \\
&= \mathrm{E}\left[\sum_{i=0}^{n-1} e_i^2(\omega)(t_{i+1} - t_i) \right] \\
&= \mathrm{E}\left[\int_0^T X_t(\omega)^2 dt \right]
\end{aligned}
$$

which is what we had to show. □

Theorem 1.4.2. *(Martingale Property of Stochastic Integrals) The stochastic integral,* $Y_t := \int_0^t X_s(\omega) dW_s(\omega)$*, is a martingale for any* $X_t(\omega) \in L^2[0,T]$.

Example 1.4.2. *Check that* $\int_0^t X_s(\omega) dW_t(\omega)$ *is indeed a martingale when* X_t *is an elementary process.*

1.5 Stochastic Differential Equations

Definition 1.5.1. *An* n*-dimensional Itô process,* X_t*, is a process that can be represented as*

$$X_t = X_0 + \int_0^t a_s ds + \int_0^t b_s dW_s \tag{4}$$

where W *is an* m*-dimensional standard Brownian motion, and* a *and* b *are* n*-dimensional and* $n \times m$*-dimensional* $\mathcal{F}_t$*-adapted processes, respectively.*

We often use the notation

$$dX_t = a_t dt + b_t dW_t$$

as shorthand for (4). An n-dimensional *stochastic differential equation* (SDE) has the form

$$dX_t = a(X_t, t)dt + b(X_t, t)dW_t; \quad X_0 = x \tag{5}$$

where as before, W_t is an m-dimensional standard Brownian motion, and a and b are n-dimensional and $n \times m$-dimensional adapted processes, respectively. Once again, (5) is shorthand for

$$X_t = x + \int_0^t a(X_s, s)dt + \int_0^t b(X_s, t)dW_s. \tag{6}$$

While we do not discuss the issue here, various conditions exist to guarantee existence and uniqueness of solutions to (6). A useful tool for solving SDE's is Itô's Lemma which we now discuss.

1.6 Itô's Lemma

Itô's Lemma is the most important result in stochastic calculus, the "sine qua non" of the field. We first state and give an outline proof of a basic form of the result.

Theorem 1.6.1. *(Itô's Lemma for 1-dimensional Brownian Motion) Let W_t be a Brownian motion on $[0,T]$ and suppose $f(x)$ is a twice continuously differentiable function on* **R**. *Then for any $t \leq T$ we have*

$$f(W_t) = f(0) + \frac{1}{2}\int_0^t f''(W_s)ds + \int_0^t f'(W_s)dW_s. \tag{7}$$

证明. (Sketch) Let $0 = t_0 < t_1 < t_2 < \ldots < t_n = t$ be a partition of $[0,t]$. Clearly

$$f(W_t) = f(0) + \sum_{i=0}^{n-1}(f(W_{t_{i+1}}) - f(W_{t_i})). \tag{8}$$

Taylor's Theorem implies

$$f(W_{t_{i+1}}) - f(W_{t_i}) = f'(W_{t_i})(W_{t_{i+1}} - W_{t_i}) + \frac{1}{2}f''(\theta_i)(W_{t_{i+1}} - W_{t_i})^2 \tag{9}$$

for some $\theta_i \in (W_{t_i}, W_{t_{i+1}})$. Substituting (9) into (8) we obtain

$$f(W_t) = f(0) + \sum_{i=0}^{n-1} f'(W_{t_i})(W_{t_{i+1}} - W_{t_i}) + \frac{1}{2}\sum_{i=0}^{n-1} f''(\theta_i)(W_{t_{i+1}} - W_{t_i})^2. \tag{10}$$

If we let $\delta := \max_i |t_{i+1} - t_i| \to 0$ then it can be shown that the terms on the right-hand-side of (10) converge to the corresponding terms on the right-hand-side of (7) as desired. (This should not be surprising as we know the quadratic variation of Brownian motion on $[0,t]$ is equal to t.) □

A more general version of Itô's Lemma can be stated for Itô processes.

Theorem 1.6.2. *(Itô's Lemma for 1-dimensional Itô process) Let X_t be a 1-dimensional Itô process satisfying the SDE*

$$dX_t = \mu_t dt + \sigma_t dW_t.$$

If $f(t,x) : [0,\infty) \times \mathbb{R} \to \mathbb{R}$ is a $C^{1,2}$ function and $Z_t := f(t, X_t)$ then

$$\begin{aligned} dZ_t &= \frac{\partial f}{\partial t}(t,X_t)dt + \frac{\partial f}{\partial x}(t,X_t)dX_t + \frac{1}{2}\frac{\partial^2 f}{\partial x^2}(t,X_t)(dX_t)^2 \\ &= \left(\frac{\partial f}{\partial t}(t,X_t) + \frac{\partial f}{\partial x}(t,X_t)\mu_t + \frac{1}{2}\frac{\partial^2 f}{\partial x^2}(t,X_t)\sigma_t^2\right)dt + \frac{\partial f}{\partial x}(t,X_t)\sigma_t dW_t. \end{aligned}$$

1.6.1 The "Box" Calculus

In the statement of Itô's Lemma, we implicitly assumed that $(dX_t)^2 = \sigma_t^2 dt$. The "box calculus" is a series of simple rules for calculating such quantities. In particular, we use the rules

$$dt \times dt = dt \times dW_t = 0 \quad \text{and} \quad dW_t \times dW_t = dt$$

when determining quantities such as $(dX_t)^2$ in the statement of Itô's Lemma above. Note that these rules are consistent with Theorem 2.2. When we have two correlated Brownian motions, $W_t^{(1)}$ and $W_t^{(2)}$, with correlation coefficient, ρ_t, then we easily obtain that $dW_t^{(1)} \times dW_t^{(2)} = \rho_t dt$. We use the box calculus for computing the quadratic variation of Itô processes.

Example 1.6.1. *Let $W_t^{(1)}$ and $W_t^{(2)}$ be two independent Brownian motions. Use Lévy's Theorem to show that*

$$W_t := \rho W_t^{(1)} + \sqrt{1-\rho^2} W_t^{(2)}$$

is also a Brownian motion for a given constant, ρ.

Example 1.6.2. *Suppose a stock price, S_t, satisfies the SDE*

$$dS_t = \mu_t S_t dt + \sigma_t S_t dW_t.$$

Then we can use the substitution, $Y_t = \log(S_t)$ and Itô's Lemma applied to the function $f(x) := \log(x)$ to obtain

$$S_t = S_0 \exp\left(\int_0^t (\mu_s - \sigma_s^2/2) ds + \int_0^t \sigma_s dW_s \right). \tag{11}$$

Note that S_t does not appear on the right-hand-side of (11) so that we have indeed solved the SDE. When $\mu_s = \mu$ and $\sigma_s = \sigma$ are constants we obtain

$$S_t = S_0 \exp\left((\mu - \sigma^2/2)t + \sigma dW_t \right) \tag{12}$$

so that $\log(S_t) \sim \mathcal{N}\left((\mu - \sigma^2/2)t, \sigma^2 t \right)$.

Example 1.6.3. *(Ornstein-Uhlenbeck Process) Let S_t be a security price and suppose $X_t = \log(S_t)$ satisfies the SDE*

$$dX_t = [-\gamma(X_t - \mu t) + \mu]dt + \sigma dW_t.$$

Then we can apply Itô's Lemma to $Y_t := \exp(\gamma t) X_t$ to obtain

$$\begin{aligned} dY_t &= \exp(\gamma t) dX_t + X_t d(\exp(\gamma t)) \\ &= \exp(\gamma t) \left([-\gamma(X_t - \mu t) + \mu]dt + \sigma dW_t \right) + X_t \gamma \exp(\gamma t) dt \\ &= \exp(\gamma t) \left([\gamma \mu t + \mu]dt + \sigma dW_t \right) \end{aligned}$$

so that

$$Y_t = Y_0 + \mu \int_0^t e^{\gamma s}(\gamma s + 1) ds + \sigma \int_0^t e^{\gamma s} dW_s \tag{13}$$

or alternatively (after simplifying the Riemann integral in (13))

$$X_t = X_0 e^{-\gamma t} + \mu t + \sigma e^{-\gamma t} \int_0^t e^{\gamma s} dW_s. \tag{14}$$

Once again, note that X_t does not appear on the right-hand-side of (14) so that we have indeed solved the SDE. We also obtain $\mathrm{E}[X_t] = X_0 e^{-\gamma t} + \mu t$ *and*

$$\begin{aligned} \mathrm{Var}(X_t) &= \mathrm{Var}\left(\sigma e^{-\gamma t} \int_0^t e^{\gamma s} dW_s \right) \\ &= \sigma^2 e^{-2\gamma t} \mathrm{E}\left[\left(\int_0^t e^{\gamma s} dW_s \right)^2 \right] \\ &= \sigma^2 e^{-2\gamma t} \int_0^t e^{2\gamma s} ds \quad \textit{(by Itô's Isometry)} \\ &= \frac{\sigma^2}{2\gamma} \left(1 - e^{-2\gamma t} \right). \end{aligned} \tag{15}$$

These moments should be compared with the corresponding moments for $\log(S_t)$ *in the previous example.*

Theorem 1.6.3. *(Itô's Lemma for n-dimensional Itô process) Let X_t be an n-dimensional Itô process satisfying the SDE*

$$dX_t = \mu_t dt + \sigma_t dW_t.$$

where $X_t \in \mathbf{R}^n$, $\mu_t \in \mathbf{R}^n$, $\sigma_t \in \mathbf{R}^{n\times m}$ and W_t is a standard m-dimensional Brownian motion. If $f(t,x) : [0,\infty) \times \mathbf{R}^n \to \mathbf{R}$ is a $C^{1,2}$ function and $Z_t := f(t, X_t)$ then

$$dZ_t = \frac{\partial f}{\partial t}(t, X_t)dt + \sum_i \frac{\partial f}{\partial x_i}(t, X_t)dX_t^{(i)} + \frac{1}{2}\sum_{i,j} \frac{\partial^2 f}{\partial x_i \partial x_j}(t, X_t)dX_t^{(i)}dX_t^{(j)}$$

where $dW_t^{(i)}dW_t^{(j)} = dtdW_t^{(i)} = 0$ for $i \neq j$ and $dW_t^{(i)}dW_t^{(i)} = dt$.

Example 1.6.4. *Let X_t and Y_t satisfy*

$$\begin{aligned} dX_t &= \mu_t^{(1)}dt + \sigma_t^{(1,1)}dW_t^{(1)} \\ dY_t &= \mu_t^{(2)}dt + \sigma_t^{(2,1)}dW_t^{(1)} + \sigma_t^{(2,2)}dW_t^{(2)} \end{aligned}$$

and define $Z_t := X_tY_t$. Apply the multi-dimensional version of Itô's Lemma to find the SDE satisfied by Z_t.

1.7 The Martingale Representation Theorem

The martingale representation theorem is an important result that is particularly useful for constructing replicating portfolios in complete financial models.

Theorem 1.7.1. *Suppose M_t is an $\mathcal{F}_t$-martingale where $\{\mathcal{F}_t\}_{t\geq 0}$ is the filtration generated by the n-dimensional standard Brownian motion, $W_t = (W_t^{(1)}, \ldots, W_t^{(n)})$. If $\mathrm{E}[M_t^2] < \infty$ for all t then there exists a unique n-dimensional adapted stochastic process, ϕ_t, such that*

$$M_t = M_0 + \int_0^t \phi_s^T dW_t \quad \text{for all } t \geq 0$$

where ϕ_s^T denotes the transpose of the vector, ϕ_s.

Example 1.7.1. *Let $F = W_T^3$ and define $M_t = \mathrm{E}_t[F]$. We will show that*

$$M_t = 3\int_0^t (T - s + W_s^2)dW_s \tag{16}$$

which is consistent with the Martingale Representation theorem. First we must calculate M_t. We do this using the independent increments property of Brownian motion and obtain

$$\begin{aligned} M_t &= \mathrm{E}_t[W_T^3] = \mathrm{E}_t[(W_T - W_t + W_t)^3] \\ &= \underbrace{\mathrm{E}_t[(W_T - W_t)^3]}_{=0} + \mathrm{E}_t[W_t^3] + 3\mathrm{E}_t[W_t(W_T - W_t)^2] + \underbrace{3\mathrm{E}_t[W_t^2(W_T - W_t)]}_{=0} \\ &= W_t^3 + 3W_t(T - t). \quad (17) \end{aligned}$$

We can now apply Itô's Lemma to (17) to obtain

$$\begin{aligned} dM_t &= 3W_t^2 dW_t + \frac{1}{2}\cdot 6W_t dt + 3(T - t)dW_t - 3W_t dt \\ &= 3(W_t^2 + T - t)dW_t \end{aligned}$$

which, noting $M_0 = 0$, is (16).

1.8 Gaussian Processes

Definition 1.8.1. *A process X_t, $t \geq 0$, is a Gaussian process if $(X_{t_1}, \ldots, X_{t_n})$ is jointly normally distributed for every n and every set of times $0 \leq t_1 \leq t_2 \leq \ldots \leq t_n$.*

If X_t is a Gaussian process, then it is determined by its *mean function*, $m(t)$, and its covariance function, $\rho(s,t)$, where

$$\begin{aligned} m(t) &= \mathrm{E}[X_t] \\ \rho(s,t) &= \mathrm{E}[(X_s - m(s))(X_t - m(t))]. \end{aligned}$$

In particular, the joint moment generating function (MGF) of $(X_{t_1}, \ldots, X_{t_n})$ is given by

$$M_{t_1,\ldots,t_n}(\theta_1, \ldots, \theta_n) = \exp\left(\theta^T m(\mathbf{t}) + \frac{1}{2}\theta^T \Sigma \theta\right) \tag{18}$$

where $m(\mathbf{t}) = (m(t_1) \ldots m(t_n))^T$ and $\Sigma_{i,j} = \rho(t_i, t_j)$.

Example 1.8.1. *(Brownian motion) Brownian motion is a Gaussian process with $m(t) = 0$ and $\rho(s,t) = \min(s,t)$ for all $s, t \geq 0$.*

Theorem 1.8.1. *(Integration of a deterministic function w.r.t. a Brownian motion) Let W_t be a Brownian motion and suppose*

$$X_t = \int_0^t \delta_s dW_s$$

where δ_s is a deterministic function. Then X_t is a Gaussian process with $m(t) = 0$ and $\rho(s,t) = \int_0^{\min(s,t)} \delta_s^2 ds$.

证明. (Sketch) (1) First use Itô's Lemma to show that

$$\mathrm{E}\left[e^{uX_t}\right] = 1 + \frac{1}{2}u^2 \int_0^t \delta_s^2 \mathrm{E}\left[e^{uX_s}\right] ds. \tag{19}$$

If we set $y_t := \mathrm{E}\left[e^{uX_t}\right]$ then we can differentiate across (19) to obtain the ODE

$$\frac{dy}{dt} = \frac{1}{2}u^2 \delta_t^2 y.$$

This is easily solved to obtain the MGF for X_t,

$$\mathrm{E}\left[e^{uX_t}\right] = \exp\left(\frac{1}{2}u^2 \int_0^t \delta_s^2 ds\right) \tag{20}$$

which, as expected, is the MGF of a normal random variable with mean 0 and variance $\int_0^t \delta_s^2 ds$.

(2) We now use (20) and similar computations to show that the joint MGF of $(X_{t_1}, \ldots, X_{t_n})$ has the form given in (18) with $m(\mathbf{t}) = 0$ and $\rho(s,t) = \int_0^{\min(s,t)} \delta_s^2 ds$. (See Shreve's *Stochastic Calculus for Finance II* for further details.) □

The next theorem again concerns Gaussian processes and is often of interest when studying short-rate term structure models.

Theorem 1.8.2. *Let W_t be a Brownian motion and suppose δ_t and ϕ_t are deterministic functions. If*

$$X_t := \int_0^t \delta_u dW_u \quad \textit{and} \quad Y_t := \int_0^t \phi_u X_u du$$

then Y_t is a Gaussian process with $m(t) = 0$ and

$$\rho(s,t) = \int_0^{\min(s,t)} \delta_v^2 \left(\int_v^s \phi_y dy \right) \left(\int_v^t \phi_y dy \right) dv.$$

证明. The proof is tedious but straightforward. (Again, see Shreve's *Stochastic Calculus for Finance II* for further details.) □

Note for example that Brownian motion with drift and the Ornstein-Uhlenbeck process are both Gaussian processes. Nonetheless, we saw in Example 5.5 that these two processes behave very differently: the Ornstein-Uhlenbeck process mean-reverts and its variance tends to a finite limit as $T \to \infty$. This is not true of Brownian motion with or without drift.

1.9 The Feynman-Kac Formula

Suppose X_t is a scalar stochastic process satisfying the SDE

$$dX_t = \mu(t, X_t)dt + \sigma(t, X_t)dW_t \tag{21}$$

where as usual $\mu(t, X_t)$ and $\sigma(t, X_t)$ are $\mathcal{F}_t$-adapted and satisfy sufficient conditions to guarantee existence of a unique solution to (21). Now consider the function, $f(x,t)$, given by

$$f(t,x) = \mathrm{E}_t^x \left[\int_t^T \phi_s^{(t)} h(X_s, s) ds + \phi_T^{(t)} g(X_T) \right]$$

where

$$\phi_s^{(t)} = \exp\left(- \int_t^s r(X_u, u) du \right)$$

and the notation $\mathrm{E}_t^x[\cdot]$ implies that the expectation should be taken conditional on time t information with $X_t = x$. Note that $f(x,t)$ may be interpreted as the time t price of a security that pays dividends at a continuous rate, $h(X_s, s)$ for $s \geq t$, and with a terminal payoff $g(X_T)$ at time T. Of course $\mathrm{E}[\cdot]$ should then be interpreted as an expectation under an equivalent martingale measure with the cash account as the corresponding numeraire and $r(\cdot,\cdot)$ as the instantaneous risk-free rate.

The Feynman-Kac Theorem states that $f(\cdot,\cdot)$ satisfies the following PDE

$$\frac{\partial f}{\partial t}(t, X_t) + \frac{\partial f}{\partial x}(t, X_t)\mu_t(t, X_t) + \frac{1}{2}\frac{\partial^2 f}{\partial x^2}(t, X_t)\sigma_t^2(t, X_t) - r(x,t)f(x,t) + h(x,t) = 0, \quad (x,t) \in \mathbf{R} \times [0,T)$$

$$f(x,T) = g(x), \quad x \in \mathbf{R}$$

证明. (Sketch proof) We will use the martingale property of conditional expectations which states that if $M_t := \mathrm{E}_t[F]$ where F is a given fixed random variable then M_t is a martingale. First let

$$f(t, X_t) := \mathrm{E}_t \left[\int_t^T \phi_s^{(t)} h(X_s, s) ds + \phi_T^{(t)} g(X_T) \right] \tag{22}$$

and note that the random variable inside the expectation on the right-hand-side of (22) is a function of t. This means we cannot yet apply the martingale property of conditional expectations. However by adding $\int_0^t \phi_s^{(t)} h(X_s, s)ds$ to both sides of (22) and then multiplying both sides by $\phi_t^{(0)}$ we obtain

$$\begin{aligned}\phi_t^{(0)}\left(f(t,X_t) + \int_0^t \phi_s^{(t)} h(X_s,s)ds\right) &= \mathrm{E}_t\left[\int_0^T \phi_s^{(0)} h(X_s,s)ds + \phi_T^{(0)} g(X_T)\right] \\ &= \mathrm{E}_t[Z], \quad \text{say.}\end{aligned}$$

Note that Z does not depend on t and so we can now apply the martingale property of conditional expectations to conclude that the left-hand-side is a martingale. We therefore apply Itô's Lemma to the left-hand-side and set the coefficient of dt to zero. This results in the Feynman-Kac PDE given above. □

Remark 1.9.1. *The Feynman-Kac result generalizes easily to the case where X_t is an n-dimensional Itô process driven by an m-dimensional standard Brownian motion.*

The Feynman-Kac Theorem plays an important role in financial engineering as it enables us to compute the prices of derivative securities (which can be expressed as expectations according to martingale pricing theory) by solving a PDE instead. It is worth mentioning, however, that in the development of derivatives pricing theory, the PDE approach preceded the martingale approach. Indeed by constructing a replicating strategy for a given derivative security, it can be shown *directly* that the price of the derivative security satisfies the Feynman-Kac PDE.

1.10 The Kolmogorov Equations

We assume in this section that X_t again satisfies the SDE of (21). Let

$$u(x,t) := \mathrm{E}\left[\Phi(X_s) \mid X_t = x\right]$$

for some function $\Phi(\cdot)$ and with $s > t$. Then following the same argument that we used in Section 8 we see that u satisfies

$$u_t + \mu(t,x)u_x + \frac{1}{2}\sigma^2(t,x)u_{xx} = 0 \quad \text{for } t < s \text{ with } u(x,s) = \Phi(x). \tag{23}$$

Suppose we now take

$$\Phi(x) = \begin{cases} 1, & \text{for } x \le y \\ 0, & \text{otherwise.}\end{cases}$$

so that $u(x,t) = \mathrm{P}\left(X_s \le y \mid X_t = x\right) =: \mathrm{P}(t,x;s,y)$, say. Let $p(t,x;s,y)$ be the corresponding PDF so that $p(t,x;s,y) = \partial P(t,x;s,y)/\partial y$. By (formally) differentiating across (23) with respect to y we obtain *Kolmogorov's backward equation* for the transition density, $p(t,x;s,y)$. In particular we obtain

$$p_t + \mu(t,x)p_x + \frac{1}{2}\sigma^2(t,x)p_{xx} = 0 \quad \text{for } t < s. \quad \text{(Kolmogorov's Backward Equation)} \tag{24}$$

It is worth emphasizing that (24) does not always admit a unique solution. In particular boundary conditions need to be specified. For example *absorbing* and *reflecting* Brownian motions both satisfy (24) but they

are very different processes. It is also worth mentioning that (23) itself is what people sometimes have in mind when they refer to Kolmogorov's backward equation.

In order to state *Kolmogorov's forward equation* we must now view $p(t,x;s,y)$ as a function of (s,y) and therefore keep (t,x) fixed. It can be shown that

$$p_s + (\mu(s,y)p)_y - \frac{1}{2}(\sigma^2(t,y)p)_{yy} = 0 \quad \text{for } s > t. \quad \text{(Kolmogorov's Forward Equation)} \quad (25)$$

with the initial condition $p = \delta_x(y)$ at $s = t$. The forward equation can be derived quickly in a formal manner from the backward equation using integration by parts but doing so rigorously can be very challenging. In fact, depending on technical and boundary conditions the forward equation (25) may not actually be satisfied. The forward equation plays a key role in local volatility models of security prices where it is used to derive the *Dupire formula*.

1.11 Change of Probability Measure

The majority of financial engineering models price securities using the EMM, Q, that corresponds to taking the cash account, B_t, as numeraire. Sometimes, however, it is particularly useful to work with another numeraire, N_t, and its corresponding EMM, P_N say. We now describe how to create new probability measures and how to switch back and forth between these measures.

Let Q be a given probability measure and M_t a strictly positive Q-martingale such that $\mathrm{E}_0^Q[M_t] = 1$ for all $t \in [0,T]$. We may then define a new equivalent probability measure, P^M, by defining

$$P_M(A) = \mathrm{E}^Q\left[M_T 1_A\right].$$

Note that

(i) $P_M(\Omega) = 1$

(ii) $P_M(A) \geq 0$ for every event A and

(iii) It can easily be shown that

$$P_M\left(\bigcup_i A_i\right) = \sum_i P_M\left(A_i\right)$$

whenever the A_i's form a sequence of disjoint sets.

Points (i), (ii) and (iii) above imply that P_M is indeed a probability measure. Because the *null-sets* of Q and P_M coincide we can conclude that P_M is indeed an equivalent probability measure (to Q). Expectations with respect to P_M then satisfy

$$\mathrm{E}_0^{P_M}[X] = \mathrm{E}_0^Q\left[M_T X\right]. \quad (26)$$

Example 1.11.1. *Verify (26) in the case where $X(\omega) = \sum_{i=1}^n c_i I_{\{\omega \in A_i\}}$, M_T is constant on each A_i, and where $A_1, \ldots, A_n$ form a partition of Ω.*

When we define a measure change this way, we use the notation dP_M/dQ to refer to M_T so that we often write

$$\frac{dP_M}{dQ} = M_T.$$

The following result explains how to switch between Q and P_M when we are taking conditional expectations. In particular, we have

$$\mathrm{E}_t^{P_M}[X] = \frac{\mathrm{E}_t^Q\left[\frac{dP_M}{dQ}X\right]}{\mathrm{E}_t^Q\left[\frac{dP_M}{dQ}\right]} \quad (27)$$

$$= \frac{\mathrm{E}_t^Q\left[\frac{dP_M}{dQ}X\right]}{M_t}. \quad (28)$$

Since M_t is a Q-martingale (28) follows easily from (27). We defer a proof of (27) to Appendix A as it requires the measure-theoretic definition of a conditional expectation.

We need to show that (27) is true. That is, we must show

$$\mathrm{E}_t^{P_M}[X] = \frac{\mathrm{E}_t^Q\left[\frac{dP_M}{dQ}X\right]}{\mathrm{E}_t^Q\left[\frac{dP_M}{dQ}\right]}. \quad (*)$$

To do this we first recall the definition of a conditional expectation. This definition states that Y is the P_M-expected value of X conditional on $\mathcal{F}_t$ if Y is an $\mathcal{F}_t$-measurable random variable satisfying

$$\mathrm{E}^{P_M}[\xi X] = \mathrm{E}^{P_M}[\xi Y]$$

for all bounded $\mathcal{F}_t$-measurable random variables, ξ. Y is then typically written as $\mathrm{E}^{P_M}[X \mid \mathcal{F}_t] \equiv \mathrm{E}_t^{P_M}[X]$. We are now in a position to prove (*). First note that the right-hand-side of (39) is indeed $\mathcal{F}_t$-measurable, as required. Now let ξ be $\mathcal{F}_t$-measurable. We then have

$$\begin{aligned}
\mathrm{E}^{P_M}\left[\frac{\mathrm{E}_t^Q\left[X\frac{dP_M}{dQ}\right]}{\mathrm{E}_t^Q\left[\frac{dP_M}{dQ}\right]}\xi\right] &= \mathrm{E}^Q\left[\frac{\mathrm{E}_t^Q\left[X\frac{dP_M}{dQ}\right]}{\mathrm{E}_t^Q\left[\frac{dP_M}{dQ}\right]}\xi\frac{dP_M}{dQ}\right] \\
&= \mathrm{E}^Q\left[\frac{\mathrm{E}_t^Q\left[X\frac{dP_M}{dQ}\right]}{\mathrm{E}_t^Q\left[\frac{dP_M}{dQ}\right]}\xi\mathrm{E}_t^Q\left[\frac{dP_M}{dQ}\right]\right] \\
&= \mathrm{E}^Q\left[\mathrm{E}_t^Q\left[X\frac{dP_M}{dQ}\xi\right]\right] \\
&= \mathrm{E}^Q\left[X\frac{dP_M}{dQ}\xi\right] = \mathrm{E}^{P_M}[X\xi]
\end{aligned}$$

and so (*) holds.

Example 1.11.2. *Show that if X is $\mathcal{F}_t$-measurable, i.e. X is known by time t, then $\mathrm{E}_0^{P_M}[X] = \mathrm{E}_0^Q[M_tX]$.*

Remark 1.11.1. *Since M_T is strictly positive we can set $X = I_A/M_T$ in (26) where I_A is the indicator function of the event A. We then obtain $\mathrm{E}_0^{P_M}[I_A/M_T] = \mathrm{E}_0^Q[I_A] = Q(A)$. In particular, we see that dQ/dP_M is given by $1/M_T$.*

Remark 1.11.2. *In the context of security pricing, we can take M_t to be the deflated time t price of a security with strictly positive payoff, normalized so that its expectation under Q is equal to 1. For example, let Z_t^T be the time t price of a zero-coupon bond maturing at time T, and let B_t denote the time t value of the cash account. We could then set $M_T := 1/(B_TZ_0^T)$ so that*

$$M_t = \mathrm{E}_t^Q\left[\frac{1}{B_TZ_0^T}\right]$$

is indeed a Q-martingale. The resulting measure, denoted by P_T, is sometimes called the T-forward measure. Note that we have implicitly assumed that in this context, Q refers to the EMM when we take the cash account as numeraire. We discuss P_T in further detail in Section 11.

1.12 A Useful EMM: The Forward Measure

We now discuss the *τ-forward measure*, P^τ, which is the EMM that corresponds to taking the zero-coupon bond with maturity τ as the numeraire. We therefore let Z_t^τ denote the time t price of a zero-coupon bond maturing at time $\tau \geq t$ and with face value \$1. As usual, we let Q denote the EMM corresponding to taking the cash account, B_t, as numeraire. We assume without loss of generality that $B_0 = \$1$ and now use Z_t^τ to define our new measure. To do this, set

$$\frac{dP^\tau}{dQ} = \frac{1}{B_\tau Z_0^\tau}. \tag{29}$$

Example 1.12.1. *Check that (29) does indeed define an equivalent probability measure.*

Now let C_t denote the time t price of a contingent claim that expires at time τ. We then have

$$\begin{aligned}
C_t &= B_t \mathrm{E}_t^Q\left[\frac{C_\tau}{B_\tau}\right] \quad \text{(by martingale pricing with EMM } Q\text{)} \quad (30)\\
&= \frac{B_t \mathrm{E}_t^{P^\tau}\left[\frac{C_\tau}{B_\tau} B_\tau Z_0^\tau\right]}{\mathrm{E}_t^{P^\tau}\left[B_\tau Z_0^\tau\right]} \quad \text{(going from } Q \text{ to } P^\tau \text{ using (27))} \quad (31)\\
&= \frac{B_t Z_0^\tau \mathrm{E}_t^{P^\tau}\left[C_\tau\right]}{\mathrm{E}_t^Q[1]/\mathrm{E}_t^Q[1/(B_\tau Z_0^\tau)]} \quad \text{(going from } P^\tau \text{ to } Q \text{ in the denominator of (31) using (27))}\\
&= Z_t^\tau \mathrm{E}_t^{P^\tau}\left[C_\tau\right] \quad \text{(by martingale pricing with EMM } Q\text{).} \quad (32)
\end{aligned}$$

We can now find C_t, either through equation (32) or through equation (30) where we use the cash account as numeraire. Computing C_t through (30) is our "usual method" and is often very convenient. When pricing equity derivatives, for example, we usually take interest rates, and hence the cash account, to be deterministic. This means that the factor $1/B_\tau$ in (30) can be taken outside the expectation so only the Q-distribution of C_τ is needed to compute C_t.

When interest rates are stochastic we cannot take the factor $1/B_\tau$ outside the expectation in (30) and we therefore need to find the *joint* Q-distribution of (B_τ, C_τ) in order to compute C_t. On the other hand, if we use equation (32) to compute C_t, then we only need the P^τ-distribution of C_τ, regardless of whether or not interest rates are stochastic. Working with a univariate-distribution is generally much easier than working with a bivariate-distribution so if we can easily find the P^τ-distribution of C_τ, then it can often be very advantageous to work with this distribution. The forward measure is therefore particularly useful when studying term-structure models.

Switching to a different numeraire can also be advantageous in other circumstances, even when interest rates are deterministic. For example, we will find it convenient to do so when pricing quanto-options and options on multiple underlying securities. Moreover, we will see how we can use Girsanov's Theorem to move back and forth between Q and other EMMs corresponding to other numeraires.

1.13 Girsanov's Theorem

Girsanov's Theorem is one of the most important results for financial engineering applications. When working with models driven by Brownian motions it enables us to (i) identify the equivalent martingale measure(s) corresponding to a given numeraire and (ii) to move back and forth between different EMM-numeraire pairs. Consider then the process

$$L_t := \exp\left(-\int_0^t \eta_s dW_s - \frac{1}{2}\int_0^t \eta_s^2 ds\right) \tag{33}$$

where η_s is an adapted process and W_s is a P-Brownian motion. Using Itô's Lemma we can check that $dL_t = -L_t\eta_t dW_t$ so L_t is a positive martingale with $\mathrm{E}_0^P[L_t] = 1$ for all t.

Theorem 1.13.1. *(Girsanov's Theorem) Define an equivalent probability measure, Q^η, by setting*

$$Q^\eta(A) := \mathrm{E}_0^P[L_T 1_A]. \tag{34}$$

Then $\widehat{W}_t := W_t + \int_0^t \eta_s ds$ is a standard Q^η-Brownian motion. Moreover, $\widehat{W}_t$ has the martingale representation property under Q^η.

Remark 1.13.1. *Suppose $\eta_s = \eta$, a constant. Then since $\widehat{W}_t := W_t + \eta t$ is a standard Q^η-Brownian motion, it implies that $W_t = \widehat{W}_t - \eta t$ is a Q^η-Brownian motion with drift $-\eta$.*

Example 1.13.1. *Let $dX_t = \mu_t dt + \sigma_t dW_t$ and suppose we wish to find a process, η_s, such that X_t is a Q^η-martingale. This is easily achieved as follows:*

$$\begin{aligned} dX_t &= \mu_t dt + \sigma_t dW_t \\ &= \mu_t dt + \sigma_t(d\widehat{W}_t - \eta_t dt) \\ &= \sigma_t d\widehat{W}_t \end{aligned}$$

if we set $\eta_t = \mu_t/\sigma_t$ in which case X_t is a Q^η-martingale. We can obtain some intuition for this result: suppose for example that μ_t and σ_t are both positive so that η_t is also positive. Then we can see from the definition of L_t in (33) that L_t places less weight on paths where W_t drifts upwards than it does on paths where it drifts downwards. This relative weighting adjusts X_t for the positive drift induced by μ_t with the result that under Q^η, X_t is a martingale.

Remark 1.13.2. *Note that Girsanov's Theorem enables us to compute Q^η-expectations directly without having to switch back to the original measure, P.*

We can get some additional intuition for the Girsanov Theorem by considering a random walk, $\mathbf{X} = \{X_0 = 0, X_1, \ldots, X_n\}$ with the interpretation that X_i is the value of the walk at time iT/n. In particular, X_n corresponds to the value of the random walk at time T. We assume that $\Delta X_i := X_i - X_{i-1} \sim N(0, T/n)$ under P and is independent of $X_0, \ldots, X_{i-1}$ for $i = 1, \ldots n$.

Example 1.13.2. *Suppose now that we want to compute $\theta := \mathrm{E}_0^Q[h(\mathbf{X})]$ where Q denotes the probability measure under which $\Delta X_i \sim N(\mu, T/n)$, again independently of $X_0, \ldots, X_{i-1}$. In particular, if we set $\mu := -T\eta/n$ then X_t approximates standard Brownian motion under P and X_t approximates Brownian*

motion with drift $-\eta$ under Q. If we let $f(\cdot)$ and $g(\cdot)$ denote the PDF's of $N(\mu, T/n)$ and $N(0, T/n)$ random variables, respectively, then we obtain

$$\begin{aligned}
\theta = \mathrm{E}_0^Q[h(\mathbf{X})] &= \int_{\mathbb{R}^n} h(x_1, \ldots, x_n) \left(\prod_{i=1}^n f(\Delta x_i) \right) d\Delta x_1 \ldots d\Delta x_n \\
&= \int_{\mathbb{R}^n} h(x_1, \ldots, x_n) \prod_i \left(\frac{f(\Delta x_i)}{g(\Delta x_i)} g(\Delta x_i) \right) d\Delta x_1 \ldots d\Delta x_n \\
&= \int_{\mathbb{R}^n} h(x_1, \ldots, x_n) \prod_i \left(\frac{f(\Delta x_i)}{g(\Delta x_i)} \right) \left(\prod_i g(\Delta x_i) \right) d\Delta x_1 \ldots d\Delta x_n \\
&= \mathrm{E}_0^P \left[h(X_1, \ldots, X_n) \prod_i \left(\frac{f(\Delta X_i)}{g(\Delta X_i)} \right) \right] \\
&= \mathrm{E}_0^P \left[h(X_1, \ldots, X_n) \exp\left(-\eta \sum_i \Delta X_i - \frac{\eta^2 T}{2} \right) \right] \\
&= \mathrm{E}_0^P \left[h(X_1, \ldots, X_n) \exp\left(-\eta X_n - \frac{\eta^2 T}{2} \right) \right]
\end{aligned}$$

which is consistent with our statement of Girsanov's Theorem in (33) and (34) above. (See Remark 4.)

Remark 1.13.3. ***(i)*** *As in the statement of the Girsanov Theorem itself, we could have chosen μ (and therefore η) to be adapted, i.e. to depend on prior events, in the random walk.*

(ii) *Note that Girsanov's Theorem allows the drift, but not the volatility of the Brownian motion, to change under the new measure, Q^η. It is interesting to see that we are not so constrained in the case of the random walk. Have you any intuition for why this is so?*

1.13.1 Multidimensional Girsanov's Theorem

The multidimensional version of Girsanov's Theorem is a straightforward generalization of the one-dimensional version. In particular let W_t be an n-dimensional standard P-Brownian motion and define

$$L_t := \exp\left(-\int_0^t \eta_s^T dW_s - \frac{1}{2} \int_0^t \eta_s^T \eta_s ds \right)$$

for $t \in [0, T]$. Then $\widehat{W}_t := W_t + \int_0^t \eta_s ds$ is a standard Q^η-Brownian motion where $dQ^\eta / dP = L_T$.

1.14 An Application: Martingale Pricing Theory

We now briefly outline how the martingale pricing theory that we have seen in a discrete-time, discrete-space framework translates to a continuous-time setting. We use S_t to denote the time t price of a risky asset and B_t to denote the time t value of the cash account. For ease of exposition we will initially assume that the risky asset does not pay dividends. Let $\phi_t^{(s)}$ and $\phi_t^{(b)}$ denote the number of units of the security and cash account, respectively, that is held in a portfolio at time t. Then the value of the portfolio at time t is given by

$$V_t = \phi_t^{(s)} S_t + \phi_t^{(b)} B_t.$$

Definition 1.14.1. *We say $\phi_t := (\phi_t^{(s)}, \phi_t^{(b)})$ is self-financing if*

$$dV_t = \phi_t^{(s)} dS_t + \phi_t^{(b)} dB_t.$$

Note that this definition is consistent with our definition for discrete-time models. It is also worth emphasizing the mathematical content of this definition. Note that Itô's Lemma implies

$$dV_t = \phi_t^{(s)} dS_t + \phi_t^{(b)} dB_t + \underbrace{S_t d\phi_t^{(s)} + B_t d\phi_t^{(b)}}_{A} + dS_t d\phi_t^{(s)} + dB_t d\phi_t^{(b)}$$

so that the self-financing assumption amounts to assuming that the sum of the terms in A are identically zero. Our definitions of *arbitrage*, *numeraire securities*, *equivalent martingale measures* and *complete* markets is unchanged from the discrete-time setup. We now state without proof the two fundamental theorems of asset pricing. These results mirror those from the discrete-time theory.

Theorem 1.14.1. *(The First Fundamental Theorem of Asset Pricing) There is no arbitrage if and only if there exists an EMM, Q.*

A consequence of Theorem 13.2 is that in the absence of arbitrage, the *deflated* value process, V_t/N_t, of any self-financing trading strategy is a Q-martingale. This implies that the deflated price of any attainable security can be computed as the Q-expectation of the terminal deflated value of the security.

Theorem 1.14.2. *(The Second Fundamental Theorem of Asset Pricing) Assume there exists a security with strictly positive price process and that there are no arbitrage opportunities. Then the market is complete if and only if there exists exactly one risk-neutral martingale measure, Q.*

Beyond their theoretical significance, these theorems are also important in practice. If our model is complete then we can price securities with the unique EMM that accompanies it, assuming of course the absence of arbitrage. This is the case for the Black-Scholes model as well as local volatility models. If our model is incomplete then we generally work directly under an EMM, Q, which is then calibrated to market data. In particular, the true data-generating probability measure, P, is often completely ignored when working with incomplete models. In some sense the issue of completeness then only arises when we discuss replicating or hedging strategies.

1.14.1 Determining Replicating Strategies

The following example is particularly useful in many financial engineering applications. While we focus on the case of a single risky security the result generalizes easily to multiple risky securities. We assume the market is complete of course, since only in that case can we guarantee that a replicating strategy actually exists.

Example 1.14.1. *(Wealth Dynamics and Hedging) We know the value of the cash-account, B_t, satisfies $dB_t = r_t B_t dt$ and suppose in addition that S_t satisfies*

$$dS_t = \mu_t S_t dt + \sigma_t S_t dW_t. \tag{35}$$

Then for a portfolio $(\phi_t^{(s)}, \phi_t^{(b)})$*, the portfolio value,* V_t*, at time* t *satisfies* $V_t := \phi_t^{(s)} S_t + \phi_t^{(b)} B_t$*. If* $\phi_t := (\phi_t^{(s)}, \phi_t^{(b)})$ *is self-financing, then we have*

$$\begin{aligned} dV_t &= \phi_t^{(s)} dS_t + \phi_t^{(b)} dB_t \\ &= \phi_t^{(s)} \mu_t S_t dt + \phi_t^{(s)} \sigma_t S_t dW_t + \phi_t^{(b)} r_t B_t dt \\ &= V_t \left[\frac{\phi_t^{(s)} S_t}{V_t} \mu_t + \frac{\phi_t^{(b)} B_t}{V_t} r_t \right] dt + \frac{\phi_t^{(s)} S_t}{V_t} \sigma_t V_t dW_t \\ &= V_t \left[r_t + \theta_t (\mu_t - r_t) \right] dt + \theta_t \sigma_t V_t dW_t \end{aligned}$$

where θ_t *and* $(1 - \theta_t)$ *are the fractions of time* t *wealth,* V_t*, invested in the risky asset and cash account, respectively, at time* t*.*

Suppose now that V_t *is the time* t *value of some derivative security and that we want to determine the self-financing trading strategy,* $(\phi_t^{(s)}, \phi_t^{(b)})$*, that replicates it. By using Itô's Lemma to write the dynamics of* V_t *as in (35), we can immediately determine* θ_t*, the fraction of time* t *wealth that is invested in the risky asset. In particular,* θ_t *is the coefficient of* $\sigma_t V_t dW_t$ *in the dynamics of* V_t*. Note also that we can work under any probability measure we choose since different probability measures will only change the drift of* V_t *and not the volatility. Once we have* θ_t *we immediately have* $1 - \theta_t$ *and then also* $(\phi_t^{(s)}, \phi_t^{(b)})$*.*

Remark 1.14.1. *If a security pays dividends or coupons, then in the above statements we should replace the security price with the total gain process from holding the security. For example, if a stock pays a continuous dividend yield of* q*, then* $Y_t := e^{qt} S_t$ *is the total gain process. It is what your portfolio would be worth at time* t *if you purchased one unit of the security at time* $t = 0$ *and reinvested all dividends immediately back into the stock. The portfolio* $(\phi_t^{(y)}, \phi_t^{(b)})$ *is then said to be self-financing if*

$$\begin{aligned} dV_t &= \phi_t^{(y)} dY_t + \phi_t^{(b)} dB_t \\ &= \phi_t^{(y)} d(e^{qt} S_t) + \phi_t^{(b)} dB_t \\ &= \phi_t^{(y)} \left(e^{qt} dS_t + q Y_t dt \right) + \phi_t^{(b)} dB_t \\ &= \phi_t^{(s)} \left(dS_t + q S_t dt \right) + \phi_t^{(b)} dB_t \end{aligned} \tag{36}$$

where $\phi_t^{(s)} = e^{qt} \phi_t^{(y)}$ *is the number of units of the risky asset held at time* t*. We shall make use of (36) when we price options on dividend-paying securities.*

1.14.2 The Role of Girsanov's Theorem

Suppose we again adopt the setting of Example 13.4 and that the risky security does not pay dividends. If we deflate as usual by the cash account then we can let $Z_t := S_t / B_t$ denote the deflated time t value of the risky asset. Assume also that W_t is a P-Brownian motion where P is the true data-generating probability measure. Then it is easy to check using Itô's Lemma that

$$dZ_t = Z_t(\mu_t - r)dt + Z_t \sigma_t dW_t. \tag{37}$$

But we know from the first fundamental theorem that Z_t must be a Q-martingale. Girsanov's Theorem, however, implies that Z_t is a Q-martingale only if $\widehat{W_t} := W_t + \int_0^t \eta_s ds$ is a Q-Brownian motion with

$\eta_s := (\mu_s - r)/\sigma_s$. Indeed, because there is only one such process, η_s, this Q is unique and the market is therefore complete by the second fundamental theorem. We then obtain that the Q-dynamics for S_t satisfy

$$dS_t = rS_t dt + \sigma_t S_t d\widehat{W}_t.$$

Example 1.14.2. *Suppose now that the risky asset pays a continuous dividend yield, q. Show that the Q-dynamics of S_t now satisfy*

$$dS_t = (r-q)S_t dt + \sigma_t S_t d\widehat{W}_t.$$

(Hint: Remember that it is now the deflated total gains process that is now a Q-martingale.)

We can use these observations now to derive the Black-Scholes formula. Suppose the stock-price, S_t, has P-dynamics

$$dS_t = \mu_t S_t dt + \sigma S_t dW_t$$

where W_t is a P-Brownian motion. Note that we have now assumed a constant volatility, σ, as Black and Scholes originally assumed. If the stock pays a continuous dividend yield of q then martingale pricing implies that the price of a call option on the stock with maturity T and strike K is given by

$$C_0 = \mathrm{E}_0^Q \left[e^{-rT}(S_T - K)^+ \right] \tag{38}$$

where $\log(S_T) \sim \mathcal{N}\left((r - q - \sigma^2/2)T, \sigma T\right)$. We can therefore compute the right-hand-side of (38) analytically to obtain the Black-Scholes formula.

Example 1.14.3. *Consider an equity model with two non dividend-paying securities, A and B, whose price processes, $S_t^{(a)}$ and $S_t^{(b)}$ respectively, satisfy the following SDE's*

$$\begin{aligned} dS_t^{(a)} &= rS_t^{(a)}dt + \sigma_1 S_t^{(a)} dW_t^{(1)} \\ dS_t^{(b)} &= rS_t^{(b)}dt + \sigma_2 S_t^{(b)} \left(\rho dW_t^{(1)} + \sqrt{1-\rho^2} dW_t^{(2)} \right) \end{aligned}$$

where $(W_t^{(1)}, W_t^{(2)})$ is a 2-dimensional Q-standard Brownian motion. We assume the cash account is the numeraire security corresponding to Q (which is consistent with the Q-dynamics of $S_t^{(a)}$ and $S_t^{(b)}$) and that the continuously compounded interest rate, r, is constant. Use the change of numeraire technique to compute the time 0 price, C_0 of a European option that expires at time T with payoff $\max(0, S_T^{(a)} - S_T^{(b)})$. (We will have plenty of practice with Girsanov's Theorem and changing probability measures as the course progresses.)

1.15 The Remaining Items: PDE Pricing Method

The Black-Scholes model postulates that the stock price S_t is described by a geometric Brownian motion:

$$dS_t = \alpha S_t\, dt + \sigma S_t\, dW_t^o, \quad \text{or } S_t = S_0 e^{(\alpha - \frac{1}{2}\sigma^2)t + \sigma W_t^o}.$$

Here W_t^o is a Brownian motion process defined on a probability space $(\Omega, \mathcal{F}, P^o)$. The parameter σ is the *volatility* and α is the *mean rate of growth*. The market also has a risk-free asset or "bond" with constant *interest rate* r:

$$dB_t = rB_t\, dt, \quad \text{or } B_t = B_0 e^{rt}.$$

This model originated with the work of Black, Scholes and Merton. Its shortcomings as a model of the real market have been discussed in the literature, and various modifications have been considered. Our purpose is not to decide on the most accurate model. Without intending to diminish their importance, we are *not* covering models involving multiple stocks, state-dependent or stochastic volatility, incomplete markets, regime switching, jump-diffusions, or Lévy processes. Nor are we considering model calibration or parameter estimation. Instead our purpose is to use the the simple Black-Scholes model as a "laboratory" in which to learn various techniques for calculating the prices of financial derivatives based on the no-arbitrage principle. The methods we learn here can be adapted to more complex models, though we will not pursue that.

1.15.1 No-Arbitrage and Risk-Neutral Pricing

Suppose we have a (European-style) derivative asset whose value at expiry ($t = T$) is given by a random variable X. The problem is to determine the market price V_t of this asset at times $t < T$. The absence of arbitrage in the market implies the following general *risk-neutral pricing formula*:

$$V_t = e^{rt} E[e^{-rT} X \mid \mathcal{F}_t]. \tag{1.2}$$

The critical feature here is that the expectation is *not* calculated under the original probability measure P^o but with respect to the the *risk-neutral* probability measure, which is characterized by the property that the discounted stock price,

$$e^{-rt} S_t,$$

is a martingale. We will use P for that risk-neutral probability. It will be the underlying probability measure throughout; we will have virtually no need for the original P^o.

The hypotheses on which the formula (1.2) is based are as follows.

- The description of S_t and B_t is appropriate, and the values of α, r, and σ are known.
- There are no transaction costs associated with buying and selling shares of stock or bond. Stock and bond may be held in any amounts (not limited to integral multiples or positive amounts) and can be traded continuously.
- The stock pays no dividends.
- For any time $t \leq T$ knowledge of the stock price history $\{S_u : u \leq t\}$ constitutes complete knowledge of the state of the market at time t. (I.e. there are no stochastic components of the model other than S_t. The values of all market variables at time t are prescribed by knowledge of the stock price history.) The σ-algebra $\mathcal{F}_t$ is determined (or "generated") by this information.
- X is $\mathcal{F}_T$ measurable and $E[X^2] < \infty$. (Remember, this is with respect to the risk-neutral probability P.)
- There are no arbitrage opportunities in the market. (Mathematically, this follows from the existence of the risk-neutral probability measure.)

An outline of how this leads to (1.2) is as follows. The effect of switching to the risk-neutral probability P is that the original W_t^o is no longer a Brownian motion. Instead

$$W_t = W_t^o + \left(\frac{\alpha - r}{\sigma}\right) t \tag{1.3}$$

is a Brownian motion with respect to P. The equation for the stock price process can be reformulated as

$$dS_t = rS_t\,dt + \sigma S_t\,dW_t. \tag{1.4}$$

The solution can be written explicitly:

$$S_t = S_0 e^{(r-\sigma^2/2)t+\sigma W_t}.$$

Suppose that h_t^B and h_t^S comprise a self-financing portfolio with value process V_t,

$$V_t = h_t^B B_t + h_t^S S_t$$
$$dV_t = h_t^B\,dB_t + h_t^S\,dS_t,$$

and which replicates X at time T:

$$V_T = X.$$

(This is always possible in the Black-Scholes model provided $E[X^2] < \infty$, by the completeness property and the martingale representation theorem.) Then it turns out that the discounted asset price $M_t = V_t/B_t$ is *also* a Q-martingale. This is because its stochastic differential works out to be

$$dM_t = \sigma h_t^S \frac{S_t}{B_t}\,dW_t,$$

with no dt term. By the absence of arbitrage, V_t must agree with the market value at time t of the contingent claim. The risk neutral formula (1.2) is just the statement of this martingale property:

$$M_t = E[M_T \,|\, \mathcal{F}_t], \text{ which we can rewrite as } V_t = B_t E[X/B_t \,|\, \mathcal{F}_t].$$

When the value at expiry is a function of the stock price at expiry,

$$X = \phi(S_T),$$

the market price will be given by a function $v(s,t)$ of the current time and current stock price,

$$V_t = v(S_t, t).$$

This $v(s,t)$ is called the *pricing function* for the contingent claim. The formula (1.2) reduces to a partial differential equation for v. We can see this if we use Itô's formula to derive the stochastic differential of $V_t/B_t = B_0^{-1}e^{-rt}v(S_t,t)$: we obtain

$$d[V_t/B_t] = B_t^{-1}\left(\left[v_t(S_t,t) + rsv_s(S_t,t) + \frac{1}{2}s^2\sigma^2 v_{ss}(S_t,t) - rv(S_t,t)\right]dt + v_s(S_t,t)\,dW_t\right).$$

Since $M_t = V_t/B_t$ must be a Q-martingale, the coefficient of the dt term must be 0. Thus we find that $v(s,t)$ satisfies the *Black-Scholes Partial Differential Equation*:

$$\frac{1}{2}s^2\sigma^2 v_{ss} + rsv_s - rv + v_t = 0, \quad \text{for } t < T,\ 0 < s \tag{1.5}$$

with the *terminal condition* $v(s,T) = \phi(s)$. This equation is equivalent to the martingale property of $v(S_t,t)/B_t$, and thus constitutes another expression of the risk-neutral formulation. Given $\phi(s)$ we want

to solve the PDE (1.5) to find the pricing function $v(s,t)$. The following figure illustrates the physical organization of this task.

The principle problem is to find the pricing function v. For the standard European options discussed above, this becomes the problem of solving the Black-Scholes PDE (1.5) with the given $\phi(s)$ as the terminal data. For some terminal value functions $\phi(s)$ explicit formulas for the solutions can be written down. A simple one is for a forward contract: $\phi(s) = s - K$. In that case it is easy to check that

$$v(s,t) = s - Ke^{-r(T-t)}.$$

The most famous explicit solution corresponds to a call option: $\phi(s) = (s-K)^+$. For this we get the *Black-Scholes formula*:

$$\begin{aligned} v^{\text{Call};K,T}(s,t) &= s\mathcal{N}(d^{(1)}) - Ke^{-r(T-t)}\mathcal{N}(d^{(2)}), \text{ where} \\ \mathcal{N}(y) &= \int_{-\infty}^{y} \frac{1}{\sqrt{2\pi}} e^{-u^2/2}\, du, \\ d^{(1)} &= \frac{\ln(s/K) + (r + \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}}, \\ d^{(2)} &= d^{(1)}(s,t) - \sigma\sqrt{T-t}. \end{aligned}$$

We will use $\mathcal{B}$ to denote the differential operator

$$\mathcal{B}v = \frac{1}{2}\sigma^2 s^2 v_{ss} + rsv_s - rv,$$

so that the Black-Scholes equation (1.5) is described concisely as

$$\mathcal{B}v(s,t) + \partial_t v(s,t) = 0. \tag{2.1}$$

We are interested in solving this for $0 < s$, $0 \le t \le T$, with a prescribed terminal value function

$$v(s,T) = \phi(s). \tag{2.2}$$

In this chapter we present some basic theory describing this problem. We want to explain what kind of $\phi(s)$ can be considered; how we should understand (2.2) if $\phi(s)$ is discontinuous; whether there will always be a solution; and whether there can be more than one solution for the same $\phi(s)$. We will show that (2.1) is equivalent to the classical heat equation. The questions we just raised have all been answered for the heat equation, so we can simply translate results from the literature on the heat equation into the Black-Scholes context to obtain the answers to our questions. We will see in particular that there is an integral formula for the solution v in terms of $\phi(s)$. We will end the chapter with a collection of explicit solutions and simple transformations for constructing new solutions from old ones.

1.15.2 Introduction to Partial Differential Equations

To begin we want to place the Black-Scholes equation in the standard taxonomy of partial differential equations. The Black-Scholes PDE is a *second order*, *linear*, *homogeneous* equation. It is second order because the highest partial derivative involved is of second order. It is linear because the partial derivatives

v_s, v_{ss}, v_t occur simply multiplied by known functions of the independent variables (s and t) and then added.

In general a second order linear PDE for a function $u(x,t)$ takes the form

$$au_{xx} + bu_{xt} + cu_{tt} + du_x + eu_t + fu + g = 0. \tag{2.3}$$

The coefficients are (prescribed) functions of the independent variables: $a = a(x,t), \ldots, g = g(x,t)$. When $g = 0$ the equation is homogeneous. The problem is to find the function(s) $u(x,t)$ which solve (2.3) for (x,t) in some region, usually in conjunction with some other "boundary conditions," i.e. properties or values of the function which are specified on the boundary of the region. For (2.1) the region we are interested is the vertical strip

$$0 \le t \le T, \quad 0 < s$$

and the boundary condition is the requirement that $v(s,T) = \phi(s)$ for all $s > 0$. There is no boundary condition specified for $s = 0$ (or $t = 0$); later in this chapter we will see why this is so.

Assuming that a, b, and c are not all zero (else the equation would not be second order), equations of the form (2.3) are usually categorized as one of three types: hyperbolic, parabolic, or elliptic. The best known examples of each type are the following.

- **Wave Equation** (hyperbolic): $u_{xx} - u_{tt} = 0$.
- **Heat Equation** (parabolic): $u_{xx} - u_t = 0$.
- **Laplace Equation** (elliptic): $u_{xx} + u_{yy} = 0$.

Each of the three basic PDEs above has been thoroughly studied. Other equations of the same type (hyperbolic, parabolic, or elliptic) typically share many of the same properties as the appropriate one of these examples. The Black-Scholes equation is parabolic. In fact it is fully equivalent to the heat equation by means of a change of variables which we will work out in the next section. This is very convenient for us, because every question we might ask about the Black-Scholes equation can be translated into an analogous question about the heat equation. We can then turn to the extensive literature concerning the heat equation (or parabolic PDEs in general) to find the answer.

1.15.3 Conversion of Black-Scholes to the Heat Equation

We are going to develop a change of variables which converts solutions of the Black-Scholes equation (2.1) into solutions of the classical heat equation. We will work this out through a sequence of steps.

Step 1.

First we observe that in terms of $y = \log(s)$ the terms $s\frac{\partial}{\partial s}$ and $s^2\frac{\partial^2}{\partial s^2}$ convert to constant multiples of $\frac{\partial}{\partial y}$ and $\frac{\partial^2}{\partial y^2}$. To be specific suppose

$$v(s,t) = f(y,t) = f(\log(s),t)$$

for some function $f(y,t)$. Then we find from the chain rule that

$$sv_s(s,t) = s \cdot \frac{1}{s} f_y(\log(s),t) = f_y(y,t),$$

$$s^2 v_{ss}(s,t) = s^2\left[-\frac{1}{s^2}f_y(\log(s),t) + \frac{1}{s^2}f_{yy}(\log(s),t)\right] = f_{yy}(y,t) - f_y(y,t).$$

The time derivative is unchanged: $v_t(s,t) = \partial_t[f(\log(s),t)] = f_t(y,t)$. If we make these substitutions in (2.1), we find that

$$\frac{1}{2}\sigma^2 s^2 v_{ss} + rsv_s - rv + v_t = \frac{1}{2}\sigma^2 f_{yy} + \left(r - \frac{1}{2}\sigma^2\right) f_y - rf + f_t,$$

where the left side is evaluated at (s,t) and the right side is evaluated at the corresponding value of (y,t). So in terms of the new variable $y = \log(s)$ we find that $v(s,t) = f(y,t)$ solves the Black-Scholes equation (2.1) if and only if f solves the equation

$$\frac{1}{2}\sigma^2 f_{yy} + \left(r - \frac{1}{2}\sigma^2\right) f_y + f_t - rf = 0. \tag{2.4}$$

This gives us an equation with constant coefficients, as opposed to the variable coefficients of (2.1). Also observe that $0 < s < \infty$ corresponds to $-\infty < y < \infty$.

Step 2.

We want to make additional variable changes to convert (2.4) into the heat equation. The next step is to eliminate the first order derivative f_y. This can be accomplished by a change of dependent variable:

$$f(y,t) = e^{c_1 y} g(y,t),$$

for a carefully chosen constant c_1. If we substitute this expression for f into (2.4) we obtain

$$e^{c_1 y}\left[\frac{1}{2}\sigma^2 g_{yy} + (c_1\sigma^2 + r - \frac{1}{2}\sigma^2)g_y + g_t + (c_1 - 1)(r + c_1\frac{1}{2}\sigma^2)g\right] = 0.$$

Making the choice

$$c_1 = \frac{1}{2} - \frac{r}{\sigma^2}$$

eliminates the g_y term, so that equation (2.4) for f is equivalent to the following equation for g:

$$\frac{1}{2}\sigma^2 g_{yy} + g_t + (c_1 - 1)(r + c_1\frac{1}{2}\sigma^2)g = 0. \tag{2.5}$$

Step 3.

We next eliminate the undifferentiated term g by another change of dependent variable:

$$g(y,t) = e^{c_2 t} h(y,t).$$

This makes (2.5) equivalent to

$$e^{c_2 t}\left[\frac{1}{2}\sigma^2 h_{yy} + h_t + c_2 h + (c_1 - 1)(r + c_1\frac{1}{2}\sigma^2)h\right] = 0.$$

We make the specific choice

$$c_2 = -(c_1 - 1)(r + c_1\frac{1}{2}\sigma^2) = \frac{(\sigma^2 + 2r)^2}{8\sigma^2}.$$

This makes (2.5) equivalent to

$$\frac{1}{2}\sigma^2 h_{yy} + h_t = 0. \tag{2.6}$$

Step 4.

Now we make one last change of variables to scale out the $\frac{1}{2}\sigma^2$ and reverse the sign on the time derivative. With $x = \sqrt{2}y/\sigma$, $\tau = T - t$ and

$$h(y,t) = e^{-c_2 T}u(x,\tau) = e^{-c_2 T}u(\sqrt{2}y/\sigma, T-t)$$

we find that (2.6) is equivalent to the heat equation:

$$u_{xx}(x,\tau) - u_\tau(x,\tau) = 0. \tag{2.7}$$

Now put all the steps together to obtain the conversion formula.

$$\begin{aligned} v(s,t) &= f(y,t) \\ &= e^{c_1 y}g(y,t) \\ &= e^{c_1 y + c_2 t}h(y,t) \\ &= e^{c_1 y + c_2 t}e^{-c_2 T}u(x,\tau) \\ &= e^{(c_1\sigma/\sqrt{2})x - c_2\tau}u(x,\tau). \end{aligned}$$

Note that the role of $e^{-c_2 T}$ in Step 4 is simply to get all the t occurring in the combination $T - t = \tau$. This τ is the time variable we will use for the heat equation. It is different from the t in the Black-Scholes equation.

To make this change of variables convenient to refer to, define the parameters $\mu = c_1\sigma/\sqrt{2}$ and $\nu = c_2$. (Don't confuse the Greek ν ("nu") with the Latin v.) After simplifying the expressions for μ and ν, we have

$$v(s,t) = e^{\mu x - \nu\tau}u(x,\tau), \quad \text{where} \tag{2.8}$$

$$\begin{aligned} s &= e^{\sigma x/\sqrt{2}}, \\ t &= T - \tau, \\ \mu &= \frac{1}{\sqrt{2}}\left(\frac{\sigma}{2} - \frac{r}{\sigma}\right), \\ \nu &= r + \mu^2. \end{aligned}$$

We will refer to v, s and t as *financial variables* and u, x, and τ as the corresponding *heat variables*. The above formulas establish a one-to-one conversion between functions $v(s,t) : (0,\infty)\times(-\infty,T] \to \mathbb{R}$ and functions $u(x,\tau) : (-\infty,\infty)\times[0,\infty) \to \mathbb{R}$ with the property that

$$\mathcal{B}v(s,t) + \partial_t v(s,t) = 0 \text{ if and only if } \partial_x^2 u(x,\tau) - \partial_\tau u(x,\tau) = 0.$$

For the heat equation we usually start with values of u for $\tau = 0$ prescribed by some *initial value function*,

$$u(x,0) = \psi(x),$$

and solve for u as τ moves forward in time. For the Black-Scholes equation we start with values given at $t = T$ by the terminal value function, $v(s,T) = \phi(s)$, and solve for v as t moves backwards in time. Observe that $\psi(x)$ and $\phi(s)$ are connected by our change of variables:

$$\phi(s) = v(s,T) = e^{\mu x}u(x,0) = e^{\mu x}\psi(x).$$

1.15.4 The Heat Equation: Existence and Uniqueness

We next summarize a number of important properties of the heat equation

$$u_{xx}(x,\tau) - u_\tau(x,\tau) = 0; \quad -\infty < x < \infty,\ \tau \geq 0.$$

We will translate these into financial variables to obtain analogous properties of the Black-Scholes equation. (A good first reference on PDEs in general is F. John 's PDE book. For a more extensive treatment of the heat equation specifically, see D. V. Widder's *The Heat Equation*.)

First observe that if we know two solutions $u^{(1)}$ and $u^{(2)}$ of (2.7), then any linear combination of them will also be a solution. To see this consider

$$u(x,\tau) = c_1 u^{(1)}(x,\tau) + c_2 u^{(2)}(x,\tau),$$

where c_1 and c_2 are any constants. Then

$$\begin{aligned} u_{xx} - u_\tau &= (c_1 u^{(1)}_{xx} + c_2 u^{(2)}_{xx}) - (c_1 u^{(1)}_\tau + c_2 u^{(2)}_\tau) \\ &= c_1(u^{(1)}_{xx} - u^{(1)}_\tau) + c_2(u^{(2)}_{xx} - u^{(2)}_\tau) \\ &= 0. \end{aligned}$$

This property of combining solutions is called the *principle of superposition*.

There are many solutions to the heat equation. As we said above, we will usually have an initial function $\psi(x)$ which prescribes initial data for the solution u we want. In other words, we require the solution to match ψ at $\tau = 0$:

$$u(x,0) = \psi(x) \text{ for all } x \in \mathbb{R}. \tag{2.9}$$

When we say $u(x,\tau)$ is a solution of (2.7) that obviously means that the partial derivatives u_{xx} and u_τ exist, as well as the fact that they are equal. The functions $\psi(x)$ which are of interest to us for option pricing purposes are generally *not* differentiable for every x. So we can *not* insist that $u(x,0) = \psi(x)$ be differentiable. In other words we can only expect that (2.7) holds for $\tau > 0$, not for $\tau = 0$. But we *do* want the values of $u(x,0) = \psi(x)$ to be associated with the values of $u(x,\tau)$ for $\tau > 0$. Thus what we will mean by saying that $u(x,\tau)$ is a solution of the *initial value problem* (2.7) and (2.9) is that

- $u(x,\tau)$ is continuous in $x \in \mathbb{R}$ and $\tau \geq 0$,
- $u(x,0) = \psi(x)$ for all $x \in \mathbb{R}$,
- $u(x,\tau)$ is twice continuously differentiable in $x \in \mathbb{R}$ and once differentiable in τ for $\tau > 0$,
- $u(x,\tau)$ satisfies (2.7) for all $x \in \mathbb{R}$ and $\tau > 0$.

Occasionally we need to consider $\psi(x)$ which is discontinuous at a finite number of x-values: $x_1, x_2, \ldots$. In that case we need to exclude $(x,\tau) = (x_i, 0)$ from the first two bullets.

The heat equation is intimately connected to Brownian motion. We can exploit that to find a formula for $u(x,\tau)$ in terms of $\psi(x)$. To see the connection, consider $M_t = u(\sqrt{2}W_t, T-t)$ and apply Itô's formula to see that

$$\begin{aligned} dM_t &= \sqrt{2}u_x\, dW_t + \left(\frac{1}{2}\cdot 2u_{xx} - u_\tau\right) dt \\ &= \sqrt{2}u_x\, dW_t, \end{aligned}$$

by virtue of the heat equation. So (under some integrability assumptions on u_x) M_t will be a martingale. Using that we find that

$$\begin{aligned} u(\sqrt{2}W_t, T-t) &= M_t \\ &= E[M_T \mid \mathcal{F}_t] \\ &= E[u(\sqrt{2}W_T, 0) \mid \mathcal{F}_t] \\ &= E[\psi(\sqrt{2}W_T) \mid \mathcal{F}_t]. \end{aligned} \tag{2.11}$$

Conditioned on $\sqrt{2}W_t = x$, $\sqrt{2}W_T$ is a normal random variable with mean x and variance $2(T-t)$. So, with $\tau = T - t > 0$ we can continue to work out the above as

$$\begin{aligned} u(x,\tau) &= \int_{-\infty}^{\infty} \psi(y) \frac{1}{\sqrt{2\pi \cdot 2(T-t)}} e^{-(y-x)^2/(2\cdot 2(T-t))}\, dy \\ &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{4\pi\tau}} e^{-(y-x)^2/(4\tau)} \psi(y)\, dy \\ &= \int_{-\infty}^{\infty} k(x,\tau;y)\psi(y)\, dy. \end{aligned} \tag{2.12}$$

The function

$$k(x,\tau;y) = \frac{1}{\sqrt{4\pi\tau}} e^{-(y-x)^2/(4\tau)}$$

occurring here is called the *fundamental solution* of the heat equation, or the *heat kernel*. (It's also the transition density for $\sqrt{2}W_t$.) You can check that for a fixed y, $k(x,\tau;y)$ does solve the heat equation for $\tau > 0$. According to the principle of superposition, it follows that for any choice of values y_j and constants $c_j = \psi(y_j)$ the sum

$$\sum_{j=1}^{N} \psi(y_j) k(x,\tau;y_j)$$

will also be a solution. We can view (2.12) as a limiting version of this construction. The reason (2.12) produces a solution with $u(x,0) = \psi(x)$ is that as $\tau \to 0$ the graph of $y \mapsto k(x,\tau;y)$ "bunches up" at $y = x$, so that when τ is small the integral only uses values of $\psi(y)$ for y very close to x.

The formula (2.12) is called a *convolution* because it is of the form $\int f(x-y)\psi(y)\, dy$. The formula is very nice in that it tells us how to find the solution u for any given ψ. But it is not the whole story. There are two important issues. One is that for some functions $\psi(x)$ the integral might fail to exist, particularly if $\psi(x)$ grows too fast as $x \to \pm\infty$. Secondly, it turns out that even for the nicest $\psi(x)$, there are always other solutions of the heat equation with $u(x,0) = \psi(x)$; (2.12) gives *only one* of many possible solutions! For instance, if $\psi(x) = 0$ then (2.12) produces $u(x,\tau) = 0$, but there are other solutions with $u(x,0) = 0$ which have $u(x,\tau) \neq 0$ for $\tau > 0$. So how do we know if the solution produced by (2.12) is the one we want, or if we should be looking for one of the others? What sets the solution from (2.12) apart from the others? The following theorem provides an answer to both of these issues.

Theorem 1.15.1. *Suppose $\psi(x)$ is a continuous function of $x \in \mathbb{R}$ with the property that*

$$\frac{\log(1+|\psi(x)|)}{1+|x|} \text{ is a bounded function.} \tag{2.13}$$

Then the integral in (2.12) does exist for all x and $\tau > 0$, the resulting function $u(x,\tau)$ solves the heat equation for $\tau > 0$ and is a continuous function of $x \in \mathbb{R}$ and $\tau \geq 0$ with $u(x,0) = \psi(x)$. Moreover u is the only such solution with the property that for each $T < \infty$,

$$\frac{\log(1+|u(x,\tau)|)}{1+|x|} \text{ is a bounded function of } x \in \mathbb{R},\ 0 \leq \tau \leq T. \tag{2.14}$$

We want to digest what this theorem says. Suppose K is a bound on the expression (2.14). That implies that for all $0 \leq \tau \leq T$ and all x,

$$|u(x,\tau)| \leq e^K e^{K|x|}.$$

On the other hand if there are positive constants k_1 and k_2 so that

$$|u(x,\tau)| \leq k_1 e^{k_2|x|}, \tag{2.15}$$

then

$$\begin{aligned} 1+|u(x,\tau)| &\leq (k_1+1)e^{k_2|x|} \\ &= e^{\log(1+k_1)+k_2|x|} \\ &\leq e^{K(1+|x|)}, \end{aligned}$$

where $K = \max(\log(1+k_1), k_2)$. Thus (2.14) is just a way of saying that (given T) there exist constants k_1 and k_2 for which (2.15) holds. We can describe this by saying that u *grows at most exponentially as* $|x| \to \pm\infty$. What the theorem tells us is that if $\psi(x)$ grows at most exponentially, then the integral formula (2.12) does indeed produce a solution with initial data ψ, and the solution it produces is the *only* solution which grows at most exponentially. All the other solutions *fail* to satisfy the exponential growth condition (2.14).

When $\psi(x)$ is discontinuous at a finite number of points the assertions of the theorem remain true if we are careful about the sense in which $u(x,0) = \psi(x)$. Now u will be continuous at all (x,τ) with $\tau > 0$ and at those $(x,0)$ for which ψ is continuous at x. And we can only say that $u(x,0) = \psi(x)$ at the continuity points of $\psi(x)$. With these adjustments, the theorem above remains true.

The Heat Equation on a Strip

We also need to understand the heat equation when x is limited to an interval. Suppose we want to limit x to an interval $x^{\mathrm{L}} \leq x \leq x^{\mathrm{H}}$. Then we would want to solve the heat equation on the "strip"

$$u_{xx} - u_\tau = 0 \text{ for } x^{\mathrm{L}} \leq x \leq x^{\mathrm{H}}, \quad 0 < \tau,$$

with prescribed initial values

$$u(x,0) = \psi(x) \text{ for } x^{\mathrm{L}} \leq x \leq x^{\mathrm{H}}.$$

This by itself is not enough information to determine the solution $u(x,\tau)$. In addition we need to know the values of $u(x^{\mathrm{L}},\tau)$ and $u(x^{\mathrm{H}},\tau)$ along the bottom and top of the strip. If we are given functions $u^{\mathrm{L}}(\tau)$

and $u^{\mathrm{H}}(\tau)$ then, in addition to the heat equation and initial values, we also want to ask that the solution satisfy the *boundary conditions*

$$u(x^{\mathrm{L}},\tau)=u^{\mathrm{L}}(\tau) \text{ and } u(x^{\mathrm{H}},\tau)=u^{\mathrm{H}}(\tau),\quad \tau>0. \tag{2.16}$$

If $\psi(x)$, $u^{\mathrm{L}}(\tau)$ and $u^{\mathrm{H}}(\tau)$ are continuous with agreement at the corners ($\psi(x^{\mathrm{L}})=u^{\mathrm{L}}(0)$ and $\psi(x^{\mathrm{H}})=u^{\mathrm{H}}(0)$) then there will be a unique function $u(x,\tau)$ which is continuous in the (closed) strip solving the heat equation and the prescribed initial and boundary conditions. Since x is bounded, there are no growth conditions as $|x|\to\infty$ to worry about; the boundary conditions (2.16) have essentially taken over that role. If there are a finite number of points of discontinuity for u^{L}, u^{H}, and ψ, or mismatch at the corners ($\psi(x^{\mathrm{L}})\neq u^{\mathrm{L}}(0)$ or $\psi(x^{\mathrm{H}})\neq u^{\mathrm{H}}(0)$) then u will not be continuous at those discontinuity points and will fail to satisfy the initial or boundary conditions at those particular points. However the existence and uniqueness of the solution still hold.

A formula like (2.12) which gives the solution u explicitly in terms of ψ, u^{L}, and u^{H} is possible, using either Fourier series (the method of "separation of variables") or theta-functions (see [32]). We will not write it down, because it would take us too far afield and we will not need it for any purpose below.

The Heat Equation on a Semi-Strip

The heat equation can also be considered on a semi-infinite strip like

$$-\infty<x\le x^{\mathrm{H}},\quad 0<\tau.$$

In that case a boundary condition is needed at $x=x^{\mathrm{H}}$:

$$u(x^{\mathrm{H}},\tau)=u^{\mathrm{H}}(\tau) \text{ for all } \tau>0,$$

and a growth condition (2.13) and (2.14) as $x\to-\infty$. (Or we could use the strip with $x^{\mathrm{L}}\le x$ with a boundary condition at $x=x^{\mathrm{L}}$ and a growth condition for $x\to+\infty$.) The theory of this kind of problem is analogous to that described above.

Example. Suppose we want to solve the above problem with $u^{\mathrm{H}}(\tau)=1$ and $\psi(x)=0$. We can tease a formula for the solution out of (2.12) by extending the definition of $\psi(x)$ to $x>x^{\mathrm{H}}$:

$$\psi(x)=\begin{cases}2 & \text{if } x>x^{\mathrm{H}},\\ 0 & \text{if } x\le x^{\mathrm{H}}.\end{cases}$$

Let $u(x,\tau)$ be the solution produced by (2.12).

$$u(x,\tau)=\int_{x^{\mathrm{H}}}^{\infty}2k(x,\tau;y)\,dy=2\mathcal{N}\left(\frac{x-x^{\mathrm{H}}}{\sqrt{2\tau}}\right).$$

It solves the heat equation for all $x\in\mathbb{R}$, $\tau>0$. So for $x<x^{\mathrm{H}}$ we have $u(x,0)=\psi(x)=0$. For $x=x^{\mathrm{H}}$ and $\tau>0$ we find that

$$u(x^{\mathrm{H}},\tau)=\int k(x^{\mathrm{H}},\tau;y)\psi(y)\,dy=2\int_{x^{\mathrm{H}}}^{\infty}\frac{1}{\sqrt{4\pi\tau}}e^{-(y-x^{\mathrm{H}})^2/(4\tau)}\,dy=1.$$

Thus this is the solution we sought. (There is a mismatch at the corner, so strictly speaking neither the boundary or initial conditions hold at $x = x^{\mathrm{H}}$, $\tau = 0$.)

The solution of this example has a useful interpretation for Brownian motion. Pick a value b, take $x^{\mathrm{H}} = \sqrt{2}b$ and let $u(x, \tau)$ be the solution above. We know that

$$M_t = u(\sqrt{2}W_t, T - t)$$

is a martingale. Let $\mathcal{T}^b$ be the first time the Brownian motion W_t reaches the value b. Then $M_{t\wedge\mathcal{T}^b}$ is also a martingale, so we can write

$$\begin{aligned} u(0,T) &= M_0 \\ &= E[M_{T\wedge\mathcal{T}^b}] \\ &= E[u(\sqrt{2}b, T-\mathcal{T}^b); \mathcal{T}^b \le T] + E[u(\sqrt{2}W_T, 0); T < \mathcal{T}^b] \\ &= E[1; \mathcal{T}^b \le T] + E[0; T < \mathcal{T}^b] \\ &= P(\mathcal{T}^b \le T). \end{aligned}$$

Thus $u(0,T)$ provides a formula for the distribution function of the first passage time $\mathcal{T}^b$ of Brownian motion:

$$P(\mathcal{T}^b \le T) = u(0,T) = 2\mathcal{N}\left(\frac{-\sqrt{2}b}{\sqrt{2T}}\right) = 2\mathcal{N}\left(-b/\sqrt{T}\right).$$

By differentiating with respect to $T > 0$ we calculate the density of $\mathcal{T}^b$ to be $\frac{b}{\sqrt{2\pi}}t^{-3/2}e^{-b^2/(2t)}$, for $t > 0$.

1.15.5 The Black-Scholes Equation: Existence and Uniqueness

We can now use our change of variables (2.8) to translate the results for the heat equation into analogous results for the Black-Scholes equation. Especially important is what the growth condition (2.14), or equivalently (2.15), becomes in terms of financial variables. Because $s = e^{\sigma x/\sqrt{2}}$, we see that $x \in \mathbb{R}$ corresponds to $0 < s$ and we have

$$e^{|x|} = \begin{cases} s^{\sqrt{2}/\sigma} & \text{for } s \ge 1, \\ s^{-\sqrt{2}/\sigma} & \text{for } 0 < s < 1 \end{cases} = \max(s, 1/s)^{\sqrt{2}/\sigma}.$$

Thus (2.15) means that

$$\begin{aligned} |v(s,t)| &\le s^{\sqrt{2}\mu/\sigma} e^{-\nu(T-t)} k_1 \max(s,1/s)^{k_2\sqrt{2}/\sigma} \\ &= e^{-\nu(T-t)} k_1 \begin{cases} s^{\frac{\sqrt{2}}{\sigma}(\mu+k_2)} & \text{for } s \ge 1, \\ (1/s)^{\frac{\sqrt{2}}{\sigma}(k_2-\mu)} & \text{for } 0 < s < 1. \end{cases} \end{aligned}$$

It follows that (2.15) in heat variables (for $0 \le \tau \le T$) is equivalent to saying that in financial variables there exist constants m_i so that

$$|v(s,t)| \le \begin{cases} m_1 s^{m_2} & \text{when } s \ge 1, \\ m_1 (1/s)^{m_2} & \text{when } 0 < s < 1 \end{cases} \tag{2.17}$$

for $0 \le t \le T$. We describe this by saying that v *grows at most polynomially* in s as $s \to \infty$, and in $1/s$ as $s \to 0$. Here is what our theorem about the heat equation says when translated to the Black-Scholes equation.

Theorem 1.15.2. *Suppose that $\phi(s)$ is a continuous function defined for $s > 0$ which grows at most polynomially in s as $s \to \infty$ and in $1/s$ as $s \to 0$. Then there is a unique function $v(s,t)$ defined and continuous for $0 < s$ and $0 \le t \le T$ which solves the Black-Scholes equation (1.5) for $0 < s$ and $t < T$, satisfies $v(s,T) = \phi(s)$, and which grows at most polynomially in s as $s \to \infty$ and in $1/s$ as $s \to 0$. This solution is given by the integral formula*

$$v(s,t) = e^{-r(T-t)} \int_0^\infty q(s,t;z,T)\phi(z)\,dz, \tag{2.18}$$

where

$$q(s,t;z,T) = g(T-t, \frac{z}{s})\frac{1}{s},$$
$$g(\tau,x) = \frac{1}{\sigma\sqrt{2\pi\tau}} e^{-\mu^2(T-t)} x^{-\left(1+\frac{\sqrt{2}\mu}{\sigma}+\frac{1}{2\tau\sigma^2}\log(x)\right)},$$

μ and ν being the parameters defined in (2.8).

The integral formula is simply the result of taking

$$\psi(x) = u(x,0) = e^{-\mu x} v(e^{\sigma x/\sqrt{2}}, T) = e^{-\mu x}\phi(e^{\sigma x/\sqrt{2}}),$$

using it in (2.12), and then changing back to financial variables ($y = \sqrt{2}\log(z)/\sigma$, $\psi(y) = e^{-\mu y}\phi(z)$).

$$\begin{aligned} v(s,t) &= e^{-\nu(T-t)+\mu x} \int_{-\infty}^\infty k(x,\tau;y)\psi(y)\,dy \\ &= e^{-\nu(T-t)} \int_{-\infty}^\infty k(x,\tau;y)e^{\mu x}e^{-\mu y}\phi(z)\,dy \\ &= e^{-r(T-t)} \int_0^\infty e^{-\mu^2(T-t)}k(x,\tau;y)e^{\mu(x-y)}\phi(z)\frac{\sqrt{2}}{\sigma}\frac{1}{z}\,dz \\ &= e^{-r(T-t)} \int_0^\infty q(s,t;z,T)\phi(z)\,dz. \end{aligned}$$

We obtain the formula (2.19) for $q(s,t;z,T)$ by substituting variables and simplifying the expression

$$q(s,t;z,T) = e^{-\mu^2(T-t)}\frac{\sqrt{2}}{\sigma}e^{\mu(x-y)}k(x,\tau;y)\frac{1}{z}.$$

We could arrive at this formula directly from the risk-neutral expression

$$\begin{aligned} v(s,t) &= e^{-r(T-t)}E[\phi(S_T) \mid S_t = s] \\ &= e^{-r(T-t)} \int_0^\infty q(s,t;z,T)\phi(z)\,dz, \end{aligned}$$

where q is the transition density for the stock price process, under the risk neutral probability.

The integral formula (2.18) is only of theoretical interest. Although it could be implemented numerically, it would probably be easier to convert $\phi(s)$ to $\psi(x)$; evaluate (2.12), and then convert back to financial variables. What is significant about the theorem is that, unlike the heat equation on a semi-strip,

the Black-Scholes equation *does not require boundary values* for $v(0,t)$ to be specified in order to determine a unique solution. We don't really consider $s=0$ to even be a legitimate value of the variable s; it corresponds to $x=-\infty$ in heat variables. Given $\phi(s)$, all that we require about $v(s,t)$ for $s\approx 0$ is that $v(s,t)$ is bounded by some power of $1/s$ as $s\downarrow 0$. This does *not* mean that $\lim_{s\downarrow 0} v(s,t)$ does not exist or is undetermined. It only means that we don't need values of $v(0,t)$ to be prescribed in order to determine v. In fact, as we will see in Section 3.3, if $\phi(0)=\lim_{s\downarrow 0}\phi(s)$ exists then $\lim_{s\downarrow 0} v(s,t)=e^{-r(T-t)}\phi(0)$.

Polynomial Growth and Risk-Neutral Pricing

The solutions $v(s,t)$ of (2.1) and (2.2) that we are interested in are those which come from the risk neutral pricing formula

$$v(S_t,t)=E[e^{-r(T-t)}\phi(S_T)\mid \mathcal{F}_t]. \tag{2.20}$$

We want to be sure that these *are* the solutions to which the above theorem applies, in other words that they *do* satisfy the polynomial growth conditions of the theorem. What makes this true is that all the terminal value functions $\phi(s)$ that we are interested in satisfy a linear bound:

$$|\phi(s)|\le Cs+C, \text{ for some nonnegative constant } C \text{ and all } s>0. \tag{2.21}$$

Consider some of our typical choices for $\phi(s)$.

- For a forward, $\phi(s)=s-K$, so $|\phi(s)|\le s+K$. So (2.21) holds with $C=K$.
- For a call or put, $\phi(s)=(s-K)^{\pm}$, so again $|\phi(s)|\le s+K$.
- Even for something more complicated, like a chooser ($T<T_i$):

$$\phi(s)=\max(v^{\text{Call};K_1,T_1}(s,T),v^{\text{Put};K_2,T_2}(s,T)),$$

because of the bounds

$$0\le v^{\text{Call};K_1,T_1}(s,T)=s\mathcal{N}(d_1)-K_1e^{-r(T_1-T)}\mathcal{N}(d_2)\le s,$$

and

$$0\le v^{\text{Put};K_2,T_2}(s,T)=-s[1-\mathcal{N}(d_1)]+K_2e^{-r(T_2-T)}[1-\mathcal{N}(d_2)]\le K_2,$$

we have $|\phi(s)|\le s+K_2$. Thus (2.21) holds.

If we use the linear bound (2.21) in the risk-neutral formula, we obtain

$$\begin{aligned}|v(S_t,t)|&\le e^{-r(T-t)}E^Q[|\phi(S_T)|\mid\mathcal{F}_t]\\&\le e^{-r(T-t)}E^Q[CS_T+C\mid\mathcal{F}_t]\\&=CS_t+e^{-r(T-t)}C.\end{aligned}$$

Therefore for all $t<T$, v satisfies the same linear bound as ϕ:

$$|v(s,t)|\le Cs+C.$$

This clearly satisfies the polynomial growth condition both as $s\to\infty$ and $s\to 0$. Thus the pricing functions v produced by the risk-neutral formula using $\phi(s)$ satisfying the linear growth condition (2.21) *do* correspond to solutions of the Black-Scholes equation described by our theorem above.

1.15.6 Solution Transformations

Suppose $u(x,\tau)$ solves the heat equation. It is easy to check that both

$$\tilde{u}(x,\tau) = u(x+c,\tau) \quad \text{and} \quad \tilde{u}(x,\tau) = u(-x,\tau)$$

also solve the heat equation. If we start with a solution $v(s,t)$ of the Black-Scholes equation, convert it to a solution of the heat equation using (2.8), apply one of the above, and then convert back to financial variables, we discover that the following transformations convert any solution v of the Black-Scholes equation to a new solution $\tilde{v}$:

$$\tilde{v}(s,t) = v(cs,t) \quad \text{any constant } c>0, \quad (2.22) \tag{1.1}$$

$$\tilde{v}(s,t) = s^{\kappa} v(s^{-1},t) \quad \text{where } \kappa = 1 - \frac{2r}{\sigma^2} = \sqrt{2}\sigma\mu. \quad (2.23) \tag{1.2}$$

These are verified by direct calculation.

The Basic Solutions $v^{(0)}$-$v^{(5)}$

For reference we number the solutions $v^{(i)}$ and list them along with their terminal values. The simplest ones are

$$v^{(0)}(s,t) = e^{-r(T-t)}, \quad v^{(0)}(s,T) = 1,$$
$$v^{(1)}(s,t) = s, \quad v^{(1)}(s,T) = s.$$

For the others we will let $d^{(i)}$ denote the expressions

$$d^{(1)}(s,t) = \frac{\log(s) + (r+\frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}},$$
$$d^{(2)}(s,t) = \frac{\log(s) + (r-\frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}}.$$

(Note that $d^{(1)}(s,t) = d^{(2)}(s,t) + \sigma\sqrt{T-t}$.)

$$v^{(2)}(s,t) = e^{-r(T-t)}\mathcal{N}(-d^{(2)}(s,t)), \quad v^{(2)}(s,T) = 1_{(0,1)}(s),$$
$$v^{(3)}(s,t) = e^{-r(T-t)}\mathcal{N}(d^{(2)}(s,t)), \quad v^{(3)}(s,T) = 1_{[1,\infty)}(s),$$
$$v^{(4)}(s,t) = s\mathcal{N}(-d^{(1)}(s,t)), \quad v^{(4)}(s,T) = s1_{(0,1)}(s),$$
$$v^{(5)}(s,t) = s\mathcal{N}(d^{(1)}(s,t)), \quad v^{(5)}(s,T) = s1_{[1,\infty)}(s).$$

Observe that $v^{(2)} - v^{(5)}$ all have a discontinuity in their terminal values at $s=1$. Thus the terminal conditions only hold for $s \neq 1$. These various solutions are not unrelated to each other. For instance $v^{(4)} = v^{(1)} - v^{(5)}$ and $v^{(2)} = v^{(0)} - v^{(3)}$. Notice that $d^{(i)}$ depend on r, σ and T as parameters, but we have not included that dependence explicitly in the notation.

All of these formulas are derived in the same way from (2.25) and (2.26). To illustrate, we work out $v^{(5)}$. Observe that we can write

$$S_T = S_t e^{(r-\sigma^2/2)(T-t)+\sigma\sqrt{T-t}\,Y},$$

where Y is a standard normal random variable independent of S_t (and the rest of $\mathcal{F}_t$). If we let $s = S_t$ then, taking advantage of the independence of Y from $\mathcal{F}_t$, we can write (2.20) as $v(S_t, t)$ where

$$v(s,t) = se^{-r(T-t)}E[\phi(se^{(r-\sigma^2/2)(T-t)+\sigma\sqrt{T-t}\,Y})].$$

For $v^{(5)}$ we want to use $\phi(s) = s1_{[1,\infty)}(s)$. Now

$$se^{(r-\sigma^2/2)(T-t)+\sigma\sqrt{T-t}\,Y} \geq 1$$

is equivalent to

$$Y \geq \frac{-\log(s) - (r-\sigma^2/2)(T-t)}{\sigma\sqrt{T-t}} = -d^{(2)}.$$

We can write the expression for $v^{(5)}$ in the form

$$v^{(5)} = se^{-r(T-t)} \cdot e^{(r-\sigma^2/2)(T-t)}E[e^{cY}; b \leq Y],$$

using $b = -d^{(2)}$ and $c = \sigma\sqrt{T-t}$. Now using (2.26) we obtain

$$\begin{aligned} v^{(5)} &= se^{-r(T-t)} \cdot e^{(r-\sigma^2/2)(T-t)} \cdot e^{(\sigma\sqrt{T-t})^2/2} \cdot \mathcal{N}(\sigma\sqrt{T-t} + d^{(2)}) \\ &= s\mathcal{N}(d^{(1)}). \end{aligned}$$

In case there is any doubt, one can check directly that $v^{(5)}(s,t)$ solves the Black-Scholes equation, although the calculation is somewhat tedious.